

# Getting Started With the Chroma Unreal SDK

This Chroma SDK plugin has been tested with Unreal versions 4.21 through 5.6.

## General

- The Chroma SDK allows an application or game to set the details in the Chroma Apps list within the Chroma App.
- The Chroma Plugin Unreal SDK for Chroma is available at Unreal\_ChromaSDK on Github.

This document provides a guide to integrating Chroma RGB using the Chroma Unreal SDK. Chroma can be included through premade Chroma animations or APIs. Here is the list of available methods:

- Initialize SDK: Initialize the Chroma SDK to use the library.
- Is Active: Check if the API is initialized.
- Is Connected: Check if the API is initialized.
- Play Chroma Animation: Playback a Chroma animation asset.

## Downloads

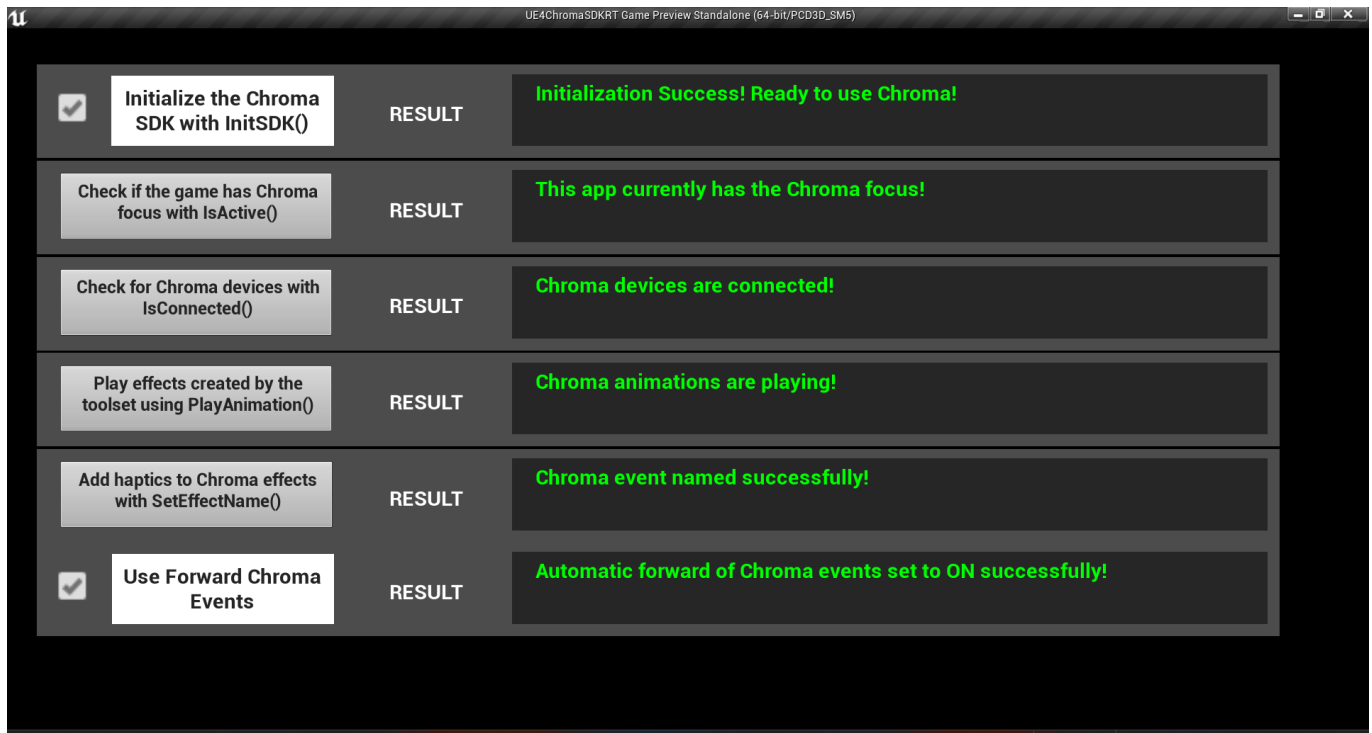
### SDKs

- CHROMA SDK

| Engine            | Git Repo            | Download Link            | Git Repo            | Download Link            | Git Repo            | Download Link            |
|-------------------|---------------------|--------------------------|---------------------|--------------------------|---------------------|--------------------------|
| Encoding          | UNICODE             |                          | ASCII               |                          | MULTI-BYTE          |                          |
| CHROMA C++ SDK    | <a href="#">Git</a> | <a href="#">Download</a> | <a href="#">Git</a> | <a href="#">Download</a> | <a href="#">Git</a> | <a href="#">Download</a> |
| Chroma Unreal SDK | <a href="#">Git</a> | <a href="#">Download</a> |                     |                          |                     |                          |
| Chroma Unity SDK  | <a href="#">Git</a> | <a href="#">Download</a> |                     |                          |                     |                          |

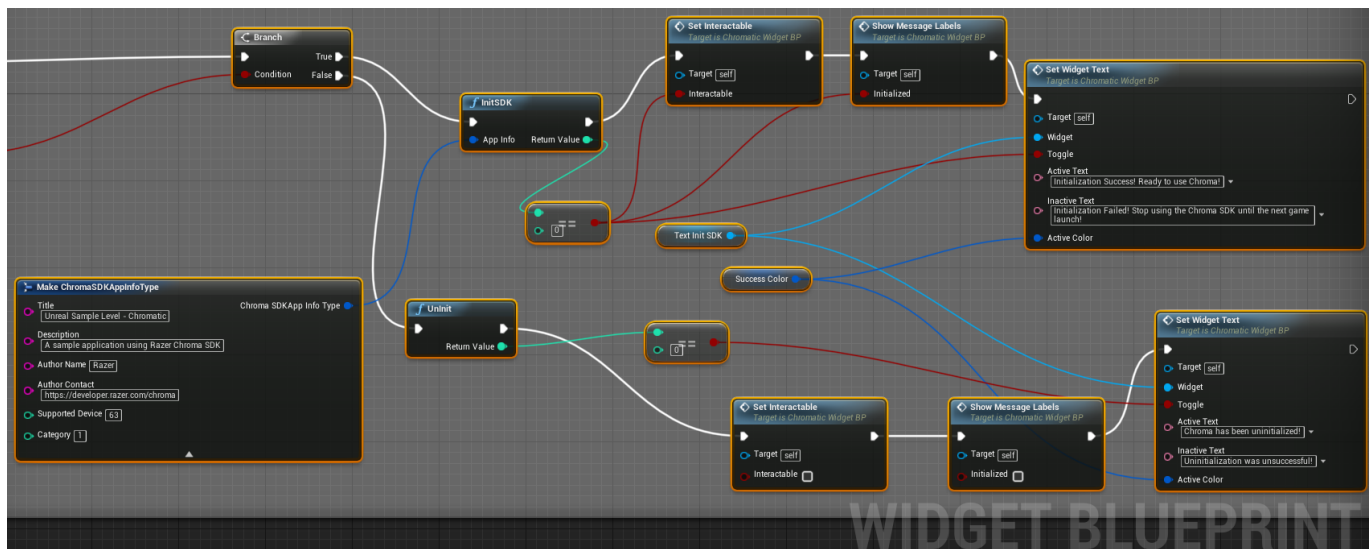
## Chromatic Level

- The following APIs are demonstrated in the Chroma\_Sample\Content\Levels\Chromatic\_Level.umap sample level and Chroma\_Sample\Content\UI\ChromaticWidget\_BP.uasset widget blueprint.



## Initialize SDK

Initialize the Chroma SDK in order to utilize the API. The `InitSDK` method takes an `AppInfo` parameter which defines the application or game details that will appear in the `Chroma App` within the `Chroma Apps` tab. The expected return result should be zero for success which indicates the API is ready for use. If a non-zero result is returned, the Chroma implementation should be disabled until the next time the application or game is launched. Reasons for failure are likely to be the user does not have the `Synapse` or the `Chroma App` installed. After successfully initializing the Chroma SDK, wait approximately 100 ms before playing Chroma animations.



```
FChromaSDKAppInfoType appInfo;
appInfo.Title = "Unreal Sample Scene - Chromatic";
appInfo.Description = "A sample application using Razer Chroma SDK";
appInfo.Author_Name = "Razer";
appInfo.Author_Contact = "https://developer.razer.com/chroma";
//appInfo.SupportedDevice =
```

```
// 0x01 | // Keyboards
// 0x02 | // Mice
// 0x04 | // Headset
// 0x08 | // Mousepads
// 0x10 | // Keypads
// 0x20 | // ChromaLink devices
appInfo.SupportedDevice = (0x01 | 0x02 | 0x04 | 0x08 | 0x10 | 0x20);
// 0x01 | // Utility. (To specify this is an utility application)
// 0x02 | // Game. (To specify this is a game);
appInfo.Category = 1;

int result = UChromaSDKPluginBPLibrary::ChromaSDKInitSDK(appInfo);
if (result == 0)
{
    // Init Success! Ready to use the Chroma SDK!
}
else
{
    // Init Failed! Stop using the Chroma SDK until the next game launch!";
}
```

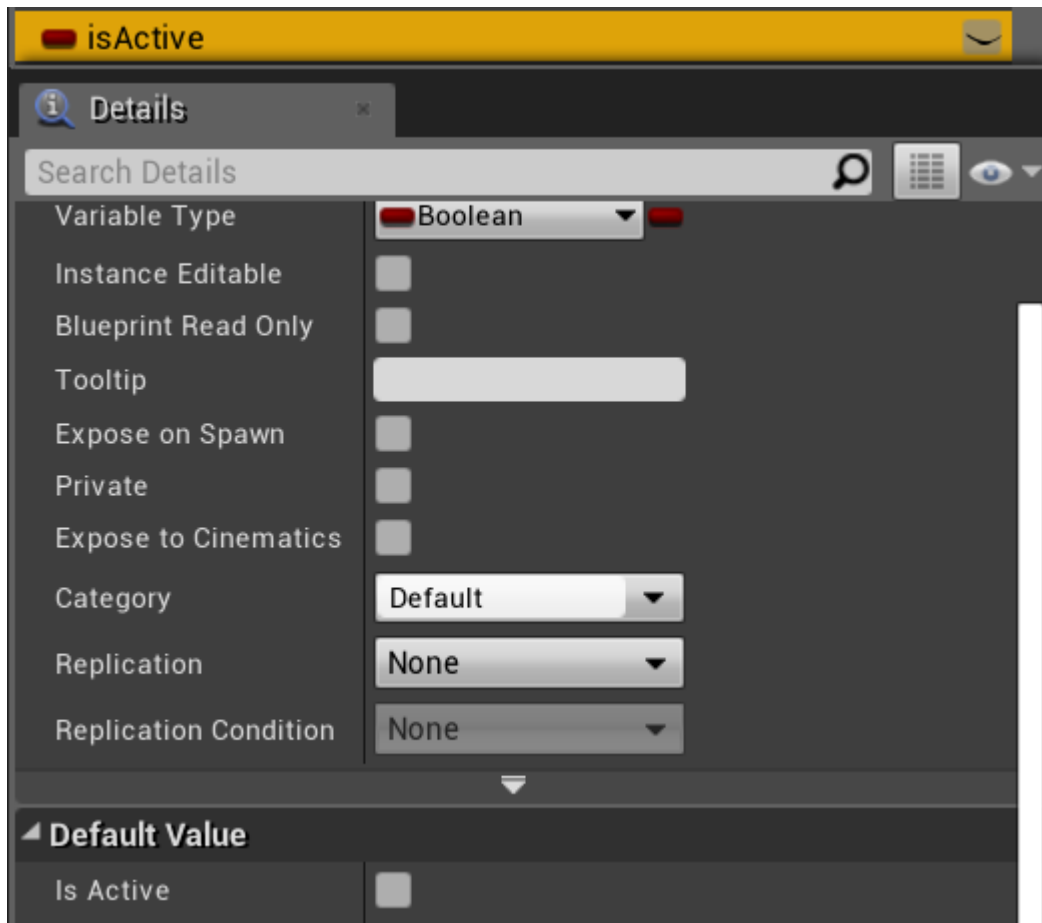
Applications should uninitialize the Chroma SDK with `Uninit()` for a clean exit. Uninitialization is only needed if the Chroma SDK was successfully initialized.

```
int result = UChromaSDKPluginBPLibrary::ChromaSDKUnInit();
if (result == 0)
{
    // Chroma has been uninitialized!
}
else
{
    // Uninitialization was unsuccessful!
}
```

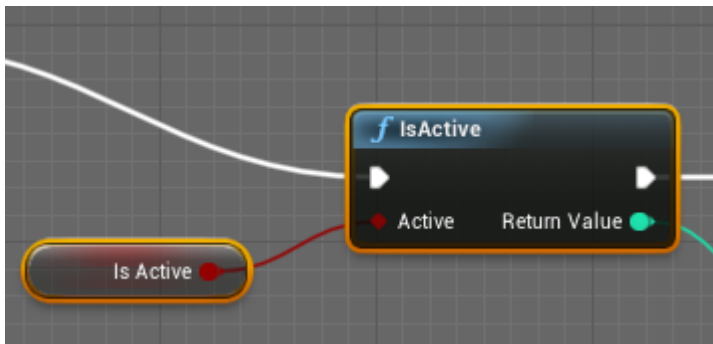
## Is Active

`IsActive()` returns the initialization state of the SDK. If the result is zero, the SDK is initialized. If the result is nonzero, avoid making calls to the API to reduce unnecessary overhead.

Blueprints should define a bool variable to pass by reference.



Check the state of the variable when the result is successful.

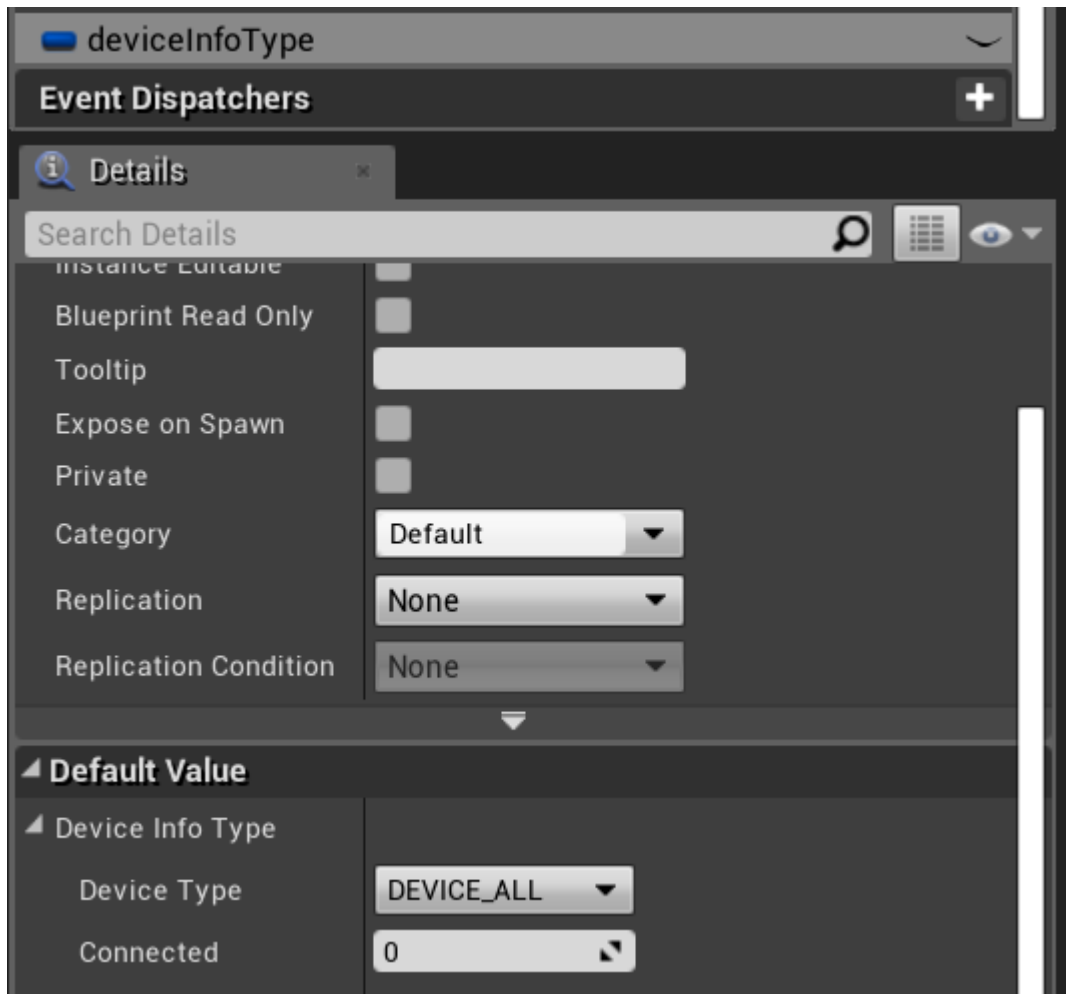


```
bool isActive;  
int result = UChromaSDKPluginBPLibrary::IsActive(isActive);  
if (result == 0)  
{  
    // The SDK is initialized  
}  
else  
{  
    // Avoid calling the API  
}
```

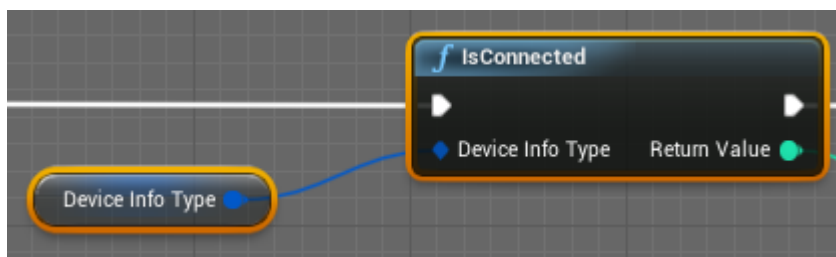
Is Connected

`IsConnected()` returns the initialization state of the SDK. If the result is zero, the SDK is initialized. If the result is nonzero, avoid making calls to the API to reduce unnecessary overhead.

Blueprints should define a `FChromaSDKDeviceInfoType` variable to pass by reference.



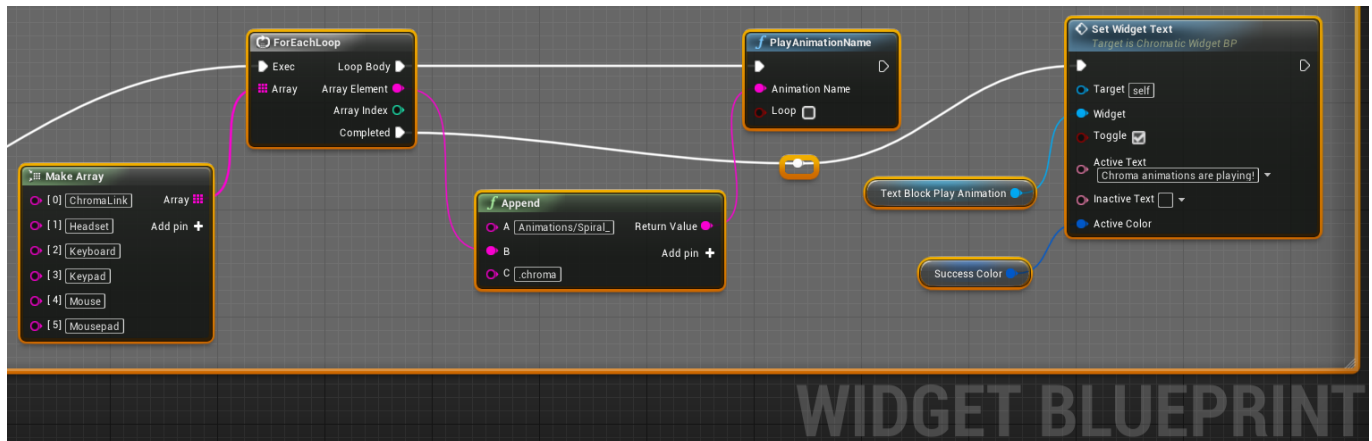
Check the state of the variable when the result is successful.



```
FChromaSDKDeviceInfoType deviceInfoType;
deviceInfoType.DeviceType = EChromaSDKCoreDeviceTypeEnum::DEVICE_ALL;
int result = UChromaSDKPluginBPLibrary::IsConnected(deviceInfoType);
if (result == 0)
{
    // The SDK is initialized
}
else
{
    // Avoid calling the API
}
```

## Play Chroma Animation

The Chroma SDK supports playing premade Chroma animations which are placed in the **Content** folder or subfolders within. Chroma animations can be created in the web authoring tools, or dynamically created and modified using the API. Call `PlayAnimation()` to play Chroma animations with or without looping. Animations have a device category, and playing an animation will stop an existing animation from playing before playing the new animation for the given device category. The animation name is file path of the Chroma animation relative to the **Content** folder.



```
bool loop = false;
TArray<FString> deviceCategories =
{
    "ChromaLink",
    "Headset",
    "Keyboard",
    "Keypad",
    "Mouse",
    "Mousepad",
};
for (int i = 0; i < deviceCategories.Num(); ++i)
{
    FString animationName = "Animations/Spiral_" + deviceCategories[i] +
    ".chroma";
    UChromaSDKPluginBPLibrary::PlayAnimationName(animationName, loop);
}
```

## Supported versions

This project is checked in under **UE 4.21**. To use a later version of Unreal, open the [Chroma\\_Sample/Chroma\\_Sample.uproject](#) project file in a text editor and change the **EngineAssociation** to the target version.

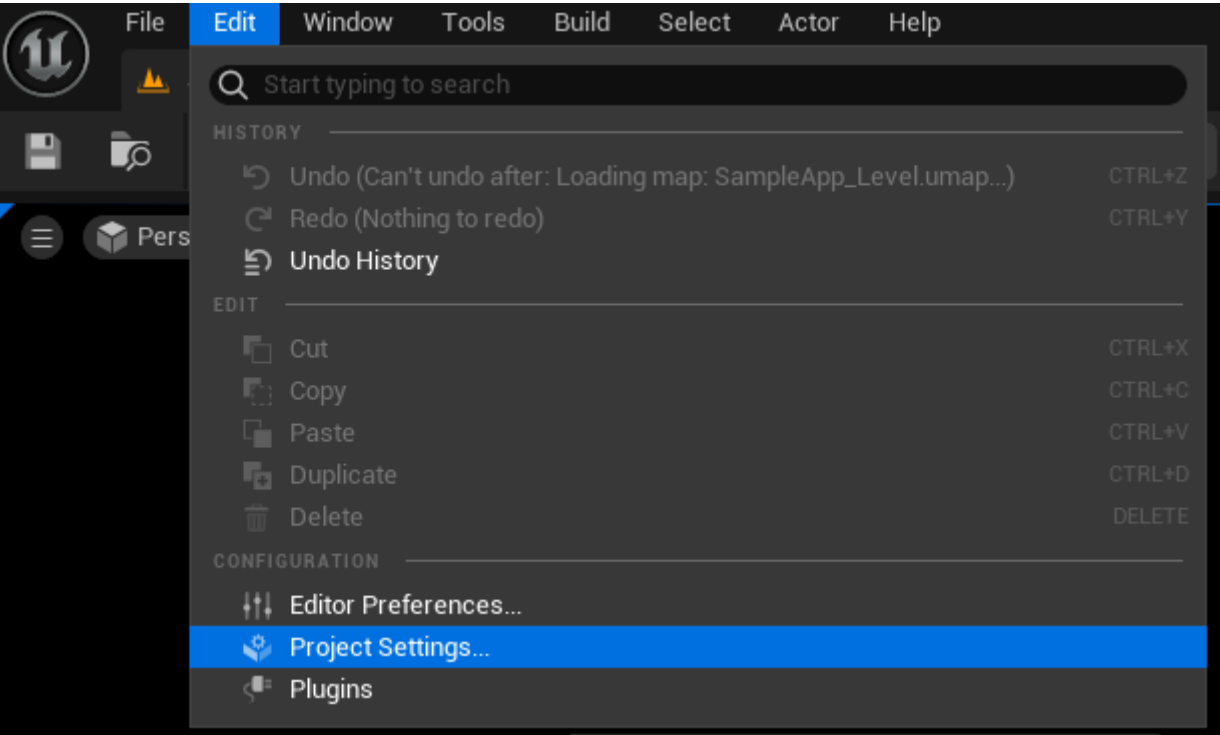
```
"EngineAssociation": "4.21",
```

To update the plugin version, open [Chroma\\_Sample/Plugins/ChromaSDKPlugin/ChromaSDKPlugin.uplugin](#) in a text editor and set the target version.

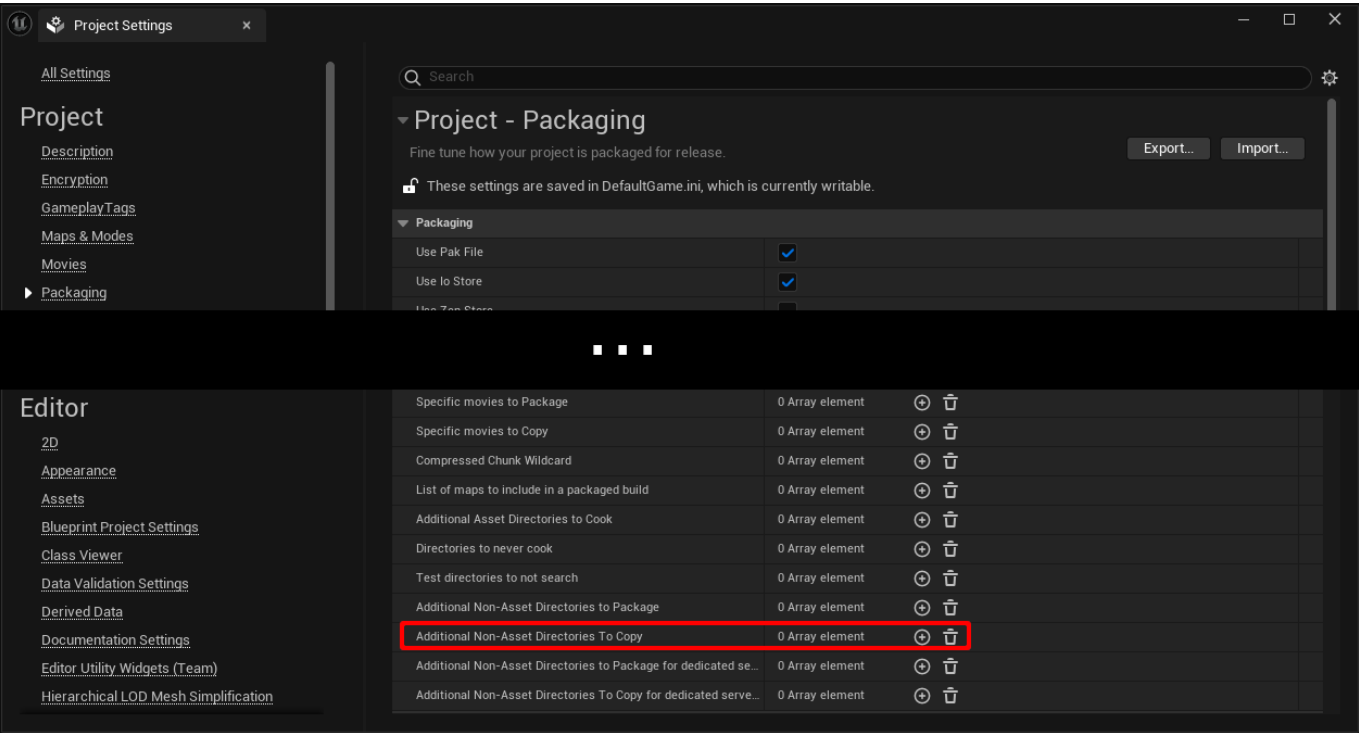
```
"EngineVersion": "4.21.0",
```

## Packaging

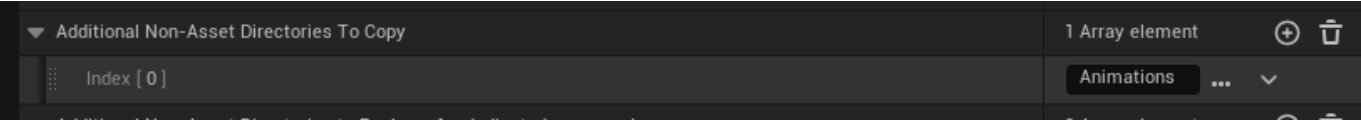
Edit the project settings in order to include Chroma animation files within the content folder.



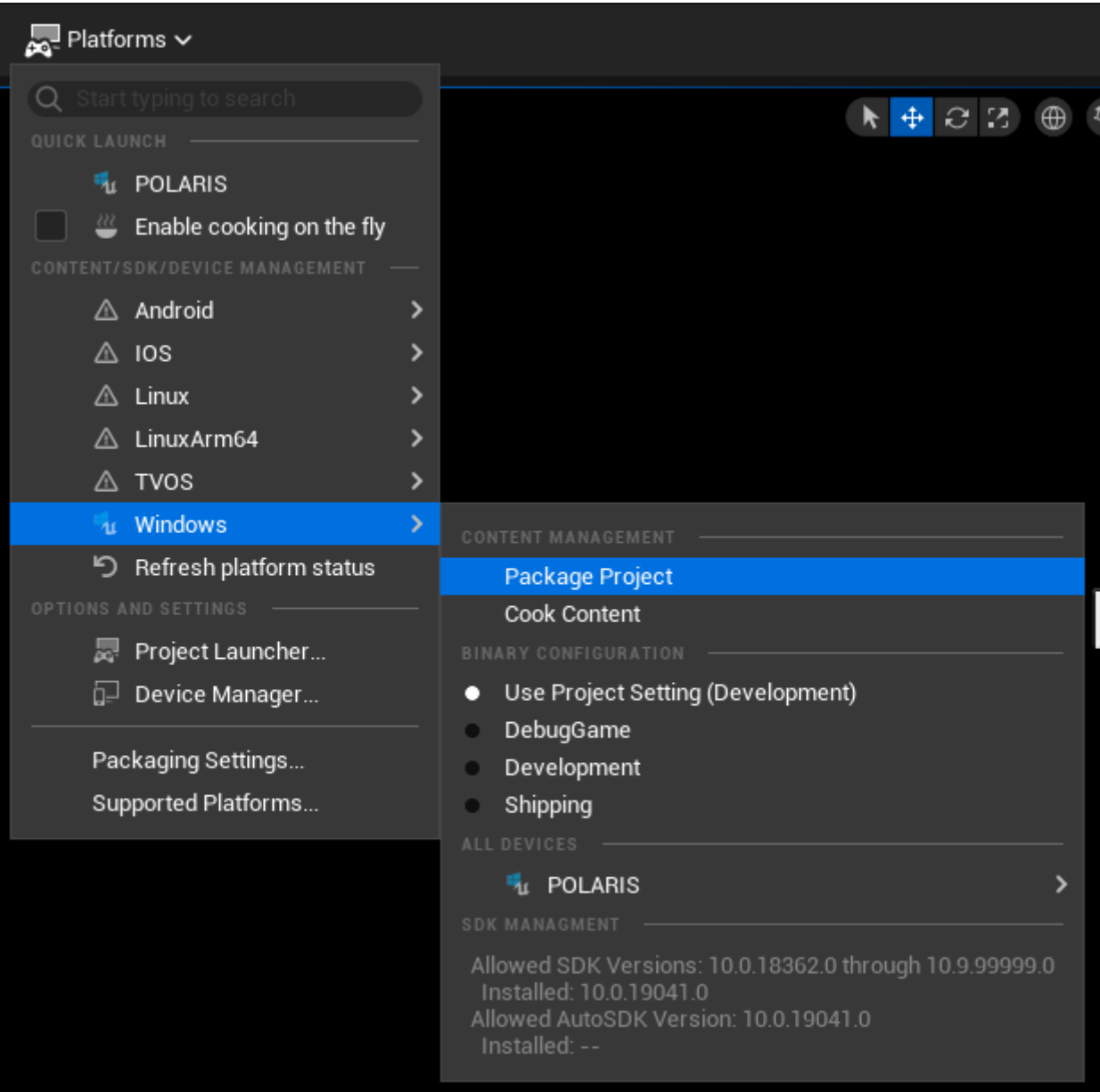
Find the Project - Packaging section and scroll down to **Additional Non-Asset Directories to Copy**.



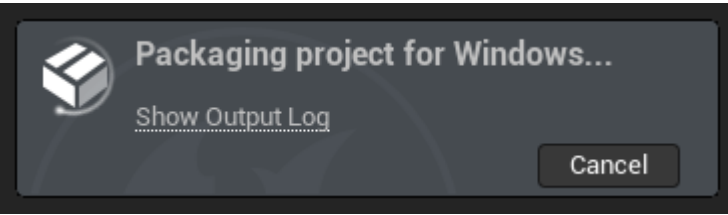
Add an entry to the location within your Content subfolder where you placed the Chroma animation files.



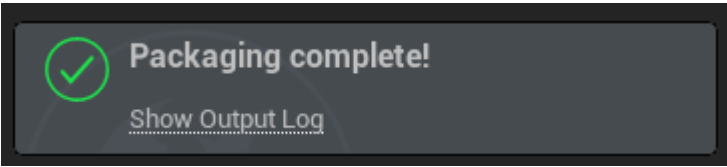
The packaging UI is different depending on your version of Unreal Editor. In 5.4, the Platforms dropdown is available on the main toolbar. Select Windows->Package Project.



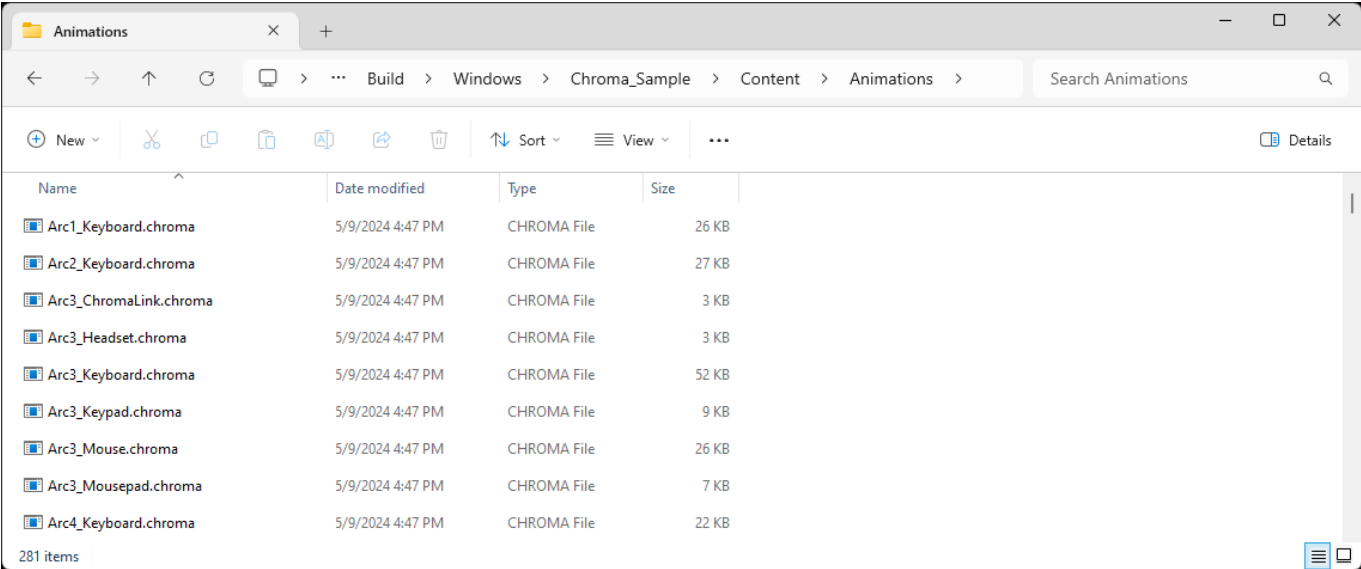
Package project for Windows... may appear for a few minutes.



And then finally complete.



After completing packaging for the Windows platforms the Chroma animation content will be included in the build automatically.



Security

To avoid a 3rd party injecting malicious code, the plugin checks for a valid signature on the Razer Chromatic Library. The DLL issuer is validated to be **Razer USA Ltd.** Init and InitSDK will return **RZRESULT\_DLL\_INVALID\_SIGNATURE** if the signature check fails.

Content

In the UE Editor, Chroma animations files are placed within the project content folder. Animation paths used in the Chroma API are relative to the content folder.

```
Chroma_Sample\Content
```

| < > ^ ⬆ ⬇ ⬅ << UE4_XDK_SampleApp > UE4ChromaSDKRT > Content > Animations ⌵ |                  |             |       |
|----------------------------------------------------------------------------|------------------|-------------|-------|
| Name                                                                       | Date modified    | Type        | Size  |
| ⌵ Arc1_Keyboard.chroma                                                     | 6/3/2022 2:52 PM | CHROMA File | 26 KB |
| ⌵ Arc2_Keyboard.chroma                                                     | 6/3/2022 2:52 PM | CHROMA File | 27 KB |
| ⌵ Arc3_ChromaLink.chroma                                                   | 6/3/2022 2:52 PM | CHROMA File | 3 KB  |
| ⌵ Arc3_Headset.chroma                                                      | 6/3/2022 2:52 PM | CHROMA File | 3 KB  |
| ⌵ Arc3_Keyboard.chroma                                                     | 6/3/2022 2:52 PM | CHROMA File | 52 KB |
| ⌵ Arc3_Keypad.chroma                                                       | 6/3/2022 2:52 PM | CHROMA File | 9 KB  |
| ⌵ Arc3_Mouse.chroma                                                        | 6/3/2022 2:52 PM | CHROMA File | 26 KB |
| ⌵ Arc3_Mousepad.chroma                                                     | 6/3/2022 2:52 PM | CHROMA File | 7 KB  |
| ⌵ Arc4_Keyboard.chroma                                                     | 6/3/2022 2:52 PM | CHROMA File | 22 KB |
| ⌵ Arrow1_Keyboard.chroma                                                   | 6/3/2022 2:52 PM | CHROMA File | 12 KB |

In a standalone PC or Cloud build, Chroma animation files may need to be copied to within the build content folder.

```
WindowsNoEditor\Chroma_Sample\Content
```

| < > ^ ⬆ ⬇ ⬅ << Build > Win64 > WindowsNoEditor > UE4ChromaSDKRT > Content > Animations |                  |             |        |
|----------------------------------------------------------------------------------------|------------------|-------------|--------|
| Name                                                                                   | Date modified    | Type        | Size   |
| ⌵ Arc1_Keyboard.chroma                                                                 | 6/3/2022 2:52 PM | CHROMA File | 26 KB  |
| ⌵ Arc2_Keyboard.chroma                                                                 | 6/3/2022 2:52 PM | CHROMA File | 27 KB  |
| ⌵ Arc3_ChromaLink.chroma                                                               | 6/3/2022 2:52 PM | CHROMA File | 3 KB   |
| ⌵ Arc3_Headset.chroma                                                                  | 6/3/2022 2:52 PM | CHROMA File | 3 KB   |
| ⌵ Arc3_Keyboard.chroma                                                                 | 6/3/2022 2:52 PM | CHROMA File | 52 KB  |
| ⌵ Arc3_Keypad.chroma                                                                   | 6/3/2022 2:52 PM | CHROMA File | 9 KB   |
| ⌵ Arc3_Mouse.chroma                                                                    | 6/3/2022 2:52 PM | CHROMA File | 26 KB  |
| ⌵ Arc3_Mousepad.chroma                                                                 | 6/3/2022 2:52 PM | CHROMA File | 7 KB   |
| ⌵ Arc4_Keyboard.chroma                                                                 | 6/3/2022 2:52 PM | CHROMA File | 22 KB  |
| ⌵ Arrow1_Keyboard.chroma                                                               | 6/3/2022 2:52 PM | CHROMA File | 12 KB  |
| ⌵ BarrelFlash1_Keyboard.chroma                                                         | 6/3/2022 2:52 PM | CHROMA File | 52 KB  |
| ⌵ BarrelFlash2_Keyboard.chroma                                                         | 6/3/2022 2:52 PM | CHROMA File | 52 KB  |
| ⌵ BarrelFlash3_Keyboard.chroma                                                         | 6/3/2022 2:52 PM | CHROMA File | 52 KB  |
| ⌵ BarRightToLeft_Keyboard.chroma                                                       | 6/3/2022 2:52 PM | CHROMA File | 13 KB  |
| ⌵ BarTopDown_Keyboard.chroma                                                           | 6/3/2022 2:52 PM | CHROMA File | 11 KB  |
| ⌵ BattleBus_Keyboard.chroma                                                            | 6/3/2022 2:52 PM | CHROMA File | 104 KB |
| ⌵ Bird1_Keyboard.chroma                                                                | 6/3/2022 2:52 PM | CHROMA File | 22 KB  |
| ⌵ BlackAndWhiteRainbow_ChromaLink.chr...                                               | 6/3/2022 2:52 PM | CHROMA File | 3 KB   |
| ⌵ BlackAndWhiteRainbow_Headset.chroma                                                  | 6/3/2022 2:52 PM | CHROMA File | 3 KB   |
| ⌵ BlackAndWhiteRainbow_Keyboard.chroma                                                 | 6/3/2022 2:52 PM | CHROMA File | 52 KB  |

## Plugin Structure

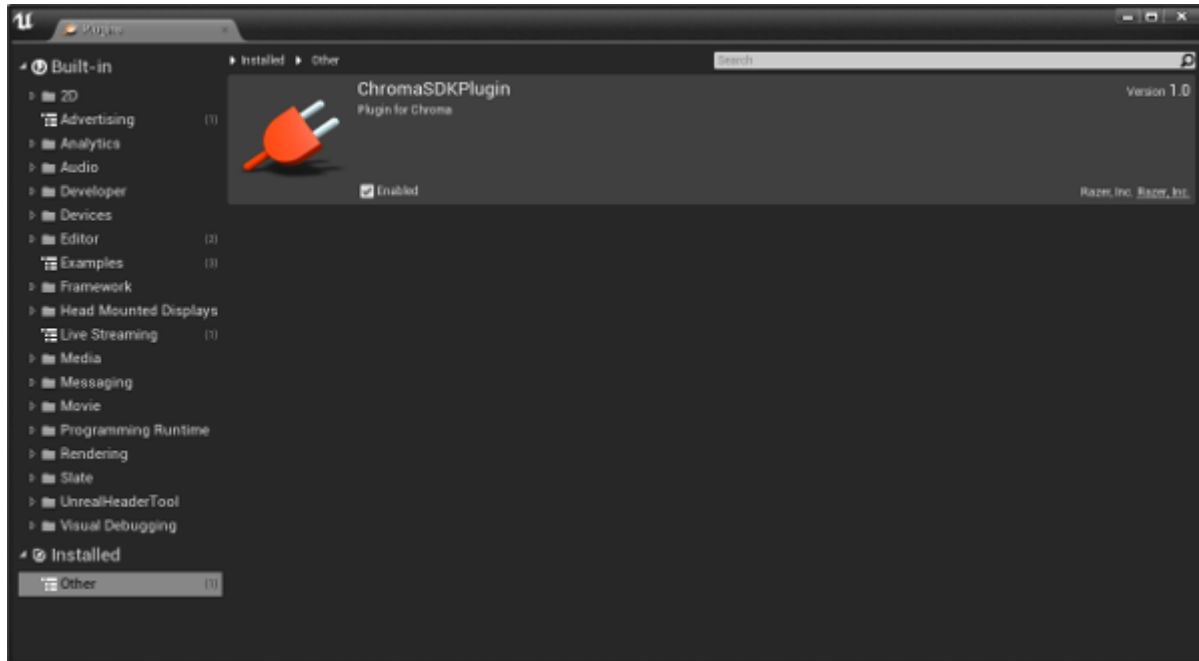
Plugin Definition: `Plugins/ChromaSDKPlugin/ChromaSDKPlugin.uplugin`

Plugin Source: `Chroma_Sample/Plugins/ChromaSDKPlugin/Source/ChromaSDKPlugin/`

Headers: `Chroma_Sample/Plugins/ChromaSDKPlugin/Source/ChromaSDKPlugin/Public/`

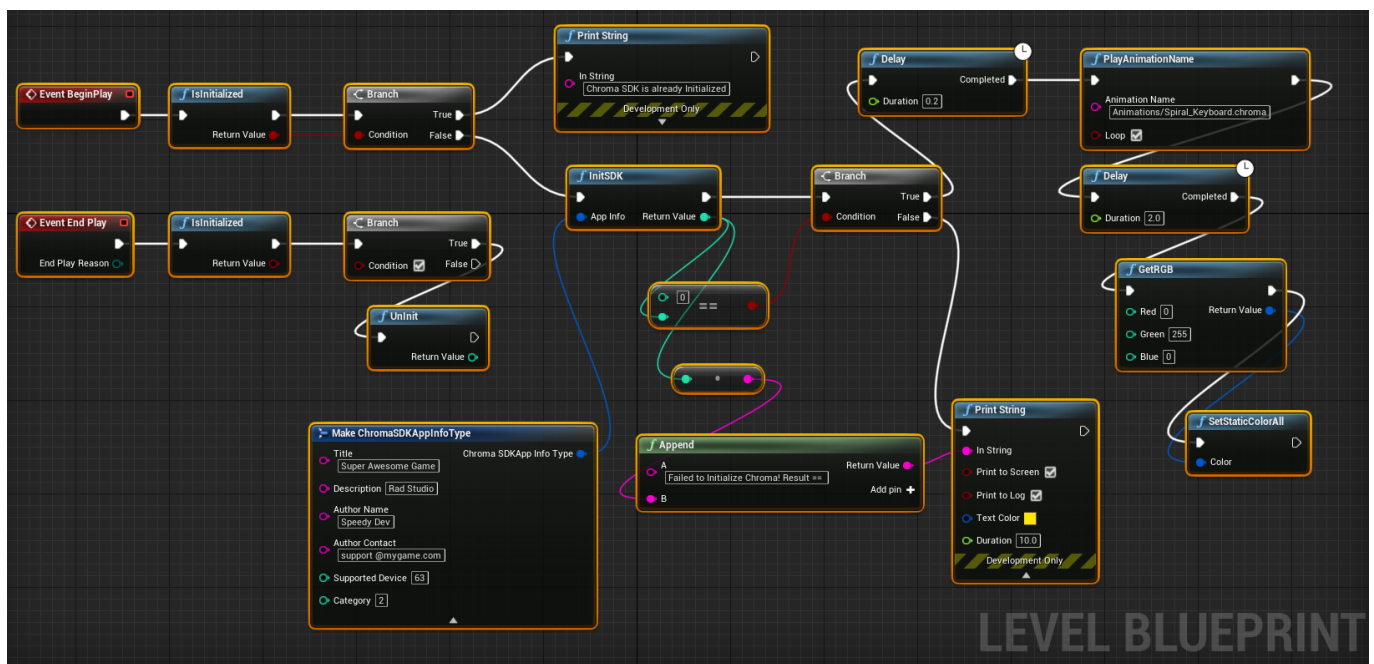
Implementation: `Chroma_Sample/Plugins/ChromaSDKPlugin/Source/ChromaSDKPlugin/Private/`

Plugin appears in Window->Plugins



## Sample Blueprint Init / Uninit Setup

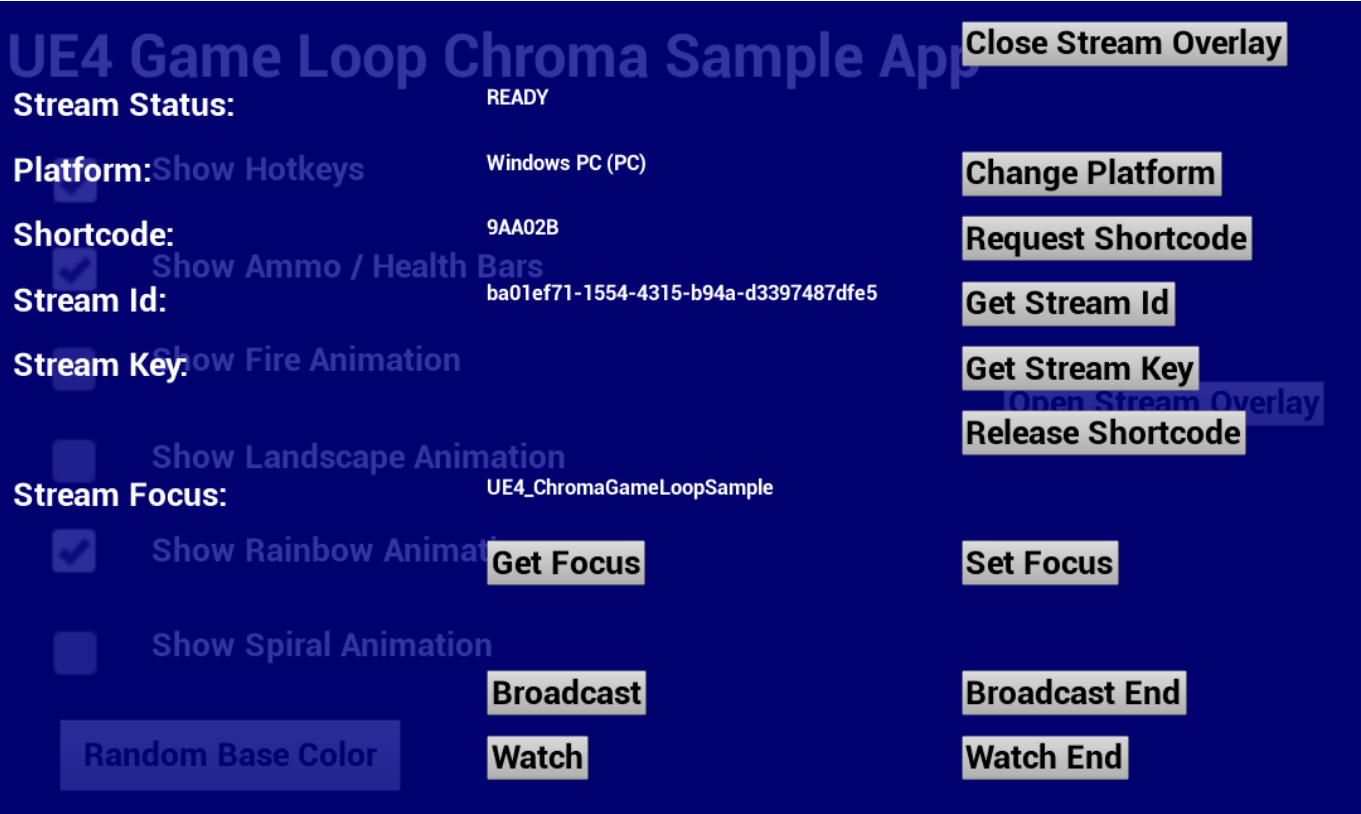
Event `BeginPlay` invokes `InitSDK` passing the `AppInfo` that provides the information that displays within `Synapse->Connect->Apps`. `InitSDK` returns `0` upon success after a 100ms delay the Chroma API is ready to use. If `InitSDK` returns nonzero, avoid further calls to the Chroma API. After success, make a call to `SupportsStreaming` and save the result. If `SupportsStreaming` returns true, the streaming API can be used for broadcasting Chroma.



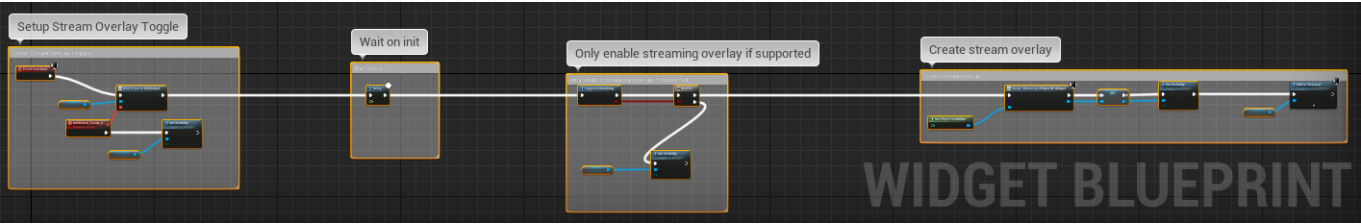
# Samples

The project has a few sample levels.

Samples share the same **Stream Overlay** logic defined in the [Chroma\\_Sample/Content/UI/StreamingWidget\\_BP.uasset](#) Widget Blueprint.

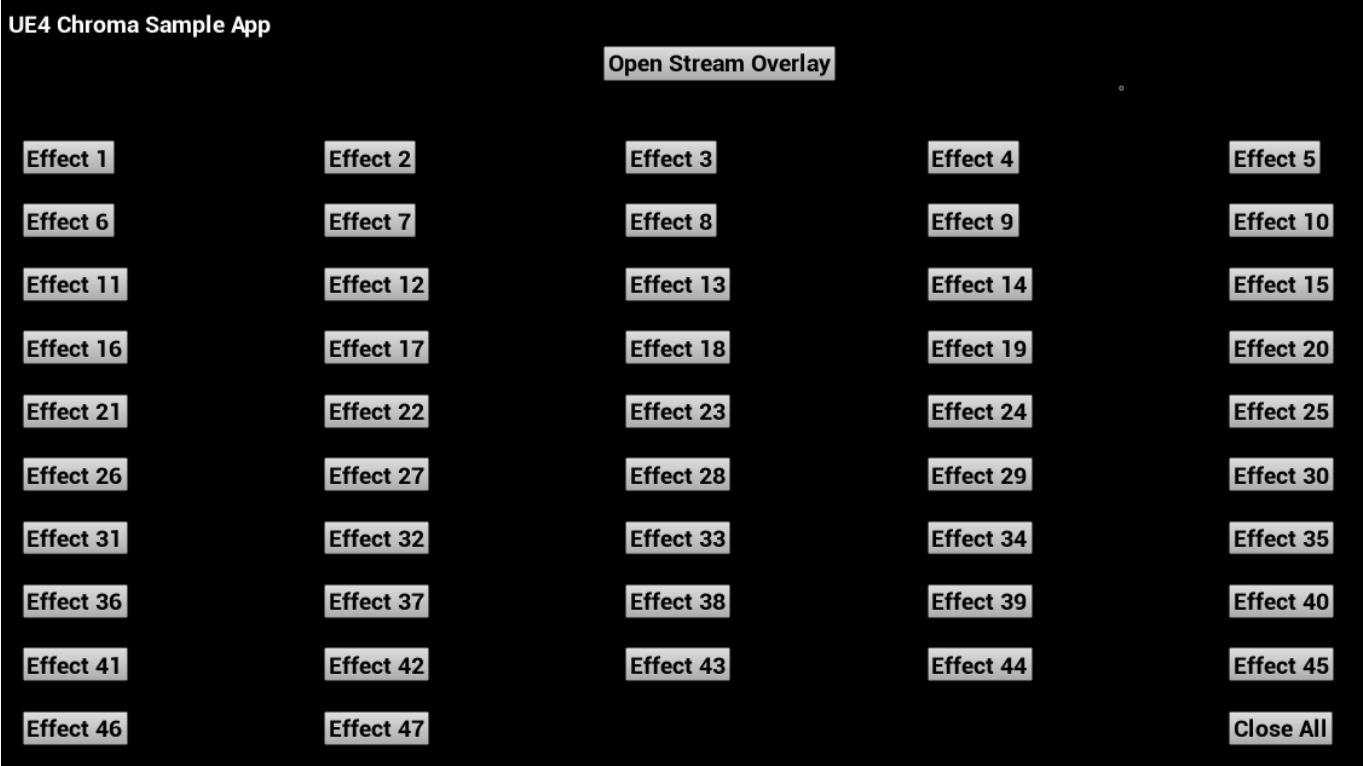


Sample UI event construction checks if streaming is supported before showing the button that displays the sample stream overlay.



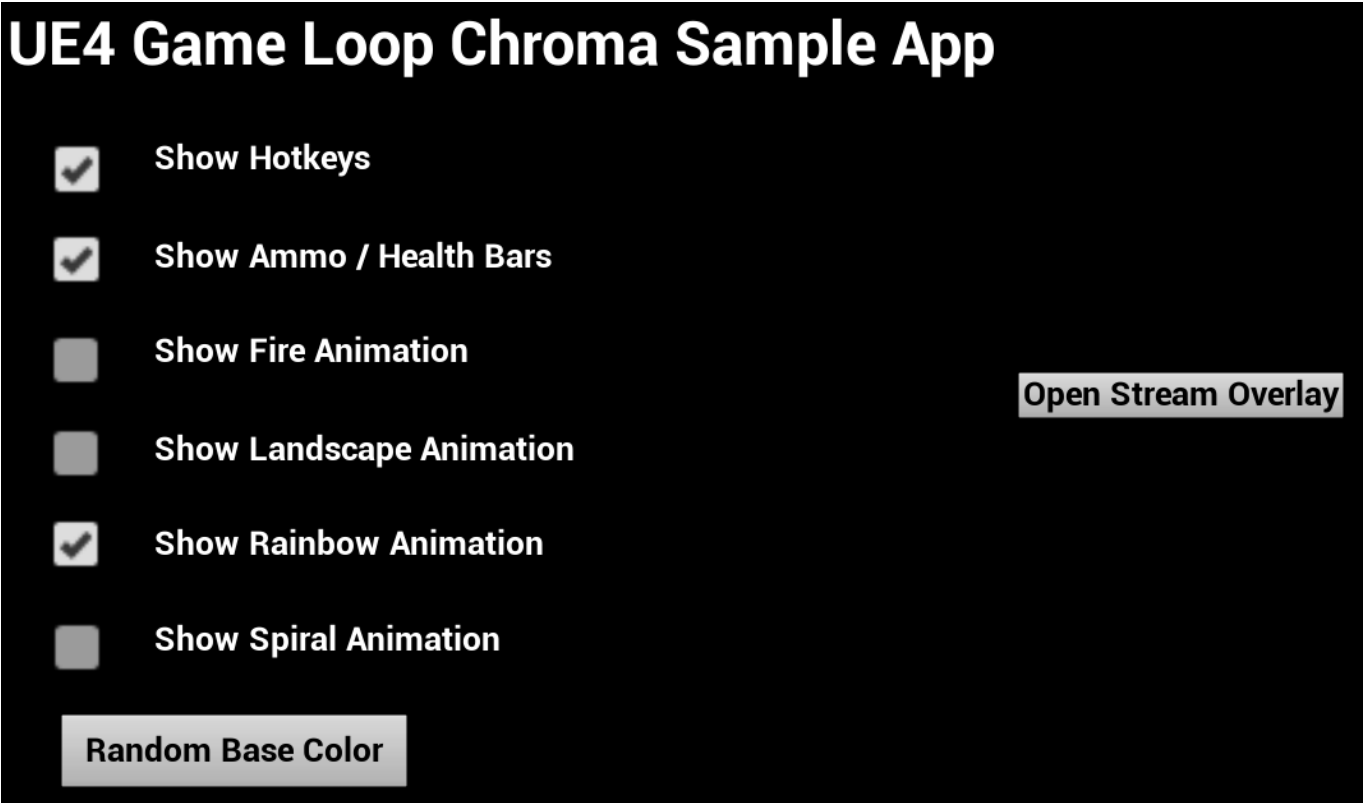
## UE Chroma Sample App

The [Chroma\\_Sample/Content/Levels/SampleApp\\_Level.umap](#) level shows the sample animations from the [Chroma Design Guide](#). The level blueprint uses BP functions defined in the [Chroma\\_Sample/Source/Chroma\\_Sample/SampleAppChromaBP.h](#) header and implemented in the [Chroma\\_Sample/Source/Chroma\\_Sample/SampleAppChromaBP.cpp](#) source.



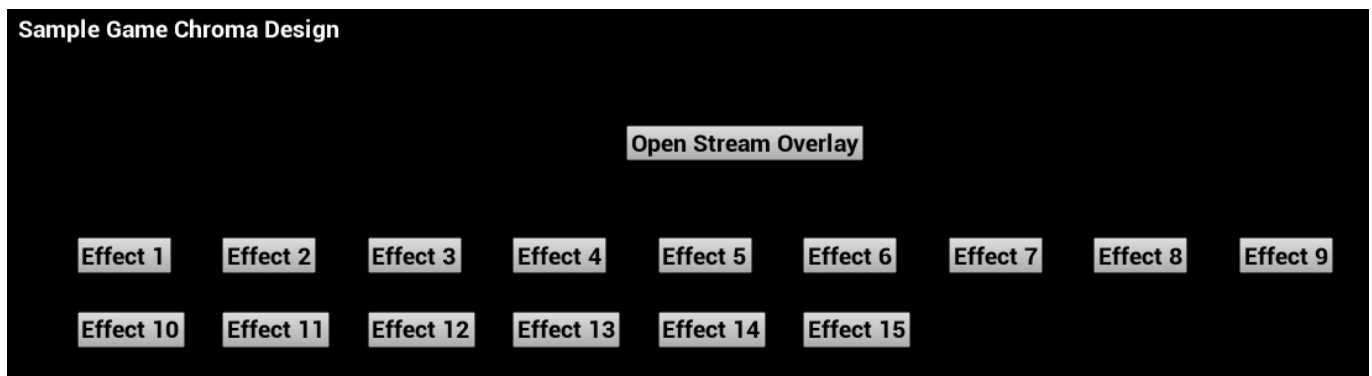
UE Game Loop Chroma Sample App

The [Chroma\\_Sample/Content/Levels/SampleGameLoopLevel.umap](#) level shows how to dynamically set color effects directly through the API and while also playing several animations at the same time using various blending operations. This sample shows how to do Chroma effects without using premade Chroma animations. Chroma animations can be used as source color information when doing dynamic blending. The level blueprint uses BP functions defined in the [Chroma\\_Sample/Source/Chroma\\_Sample/SampleGameLoopChromaBP.h](#) header and implemented in the [Chroma\\_Sample/Source/Chroma\\_Sample/SampleGameLoopChromaBP.cpp](#) source.



## UE Sample Game Chroma Design

The [Chroma\\_Sample/Content/Levels/SampleGameLevel.umap](#) level is a template intended to work with the automated [Chroma Design Converter](#) for quickly porting sample effects from HTML5 to Unity. The level blueprint uses BP functions defined in the [Chroma\\_Sample/Source/Chroma\\_Sample/SampleGameChromaBP.h](#) header and implemented in the [Chroma\\_Sample/Source/Chroma\\_Sample/SampleGameChromaBP.cpp](#) source. Chroma Design samples are commonly created with 15 sample effects which is why the template has that many buttons to play the sample effects from the ported code. The Chroma Design Converter is not limited to just 15 sample effects and can generate more effect code from the input HTML5 script.



## Unreal Compatibility

- Note: Enum syntax - Enums are namespaced and types use the `EChromaSDKKeyboardKey::Type` syntax to avoid collisions.
- Note: No const enum types or passing enums by reference in function parameters - Avoid use of const enum types because that seems to crash in UE 4.5.

## Full API

[Chroma\\_Sample](#) is a Blueprint API library with methods that expose the Chroma SDK to C++ and blueprints.

- Take a look at the code from [SampleAppChromaBP.cpp](#). These sample effects show how the blueprint library was used to create the [Chroma](#) effects that correspond to the [Guide](#).
- [AddNonZeroAllKeys](#)
- [AddNonZeroAllKeysAllFrames](#)
- [AddNonZeroAllKeysAllFramesName](#)
- [AddNonZeroAllKeysAllFramesOffset](#)
- [AddNonZeroAllKeysAllFramesOffsetName](#)
- [AddNonZeroAllKeysName](#)
- [AddNonZeroTargetAllKeysAllFrames](#)
- [AddNonZeroTargetAllKeysAllFramesName](#)
- [AddNonZeroTargetAllKeysAllFramesOffset](#)

- [AddNonZeroTargetAllKeysAllFramesOffsetName](#)
- [AppendAllFrames](#)
- [AppendAllFramesName](#)
- [ClearAll](#)
- [ClearAnimationType](#)
- [CloseAll](#)
- [CloseAnimation](#)
- [CloseAnimationName](#)
- [CopyAllKeys](#)
- [CopyAllKeysName](#)
- [CopyAnimation](#)
- [CopyAnimationName](#)
- [CopyKeyColor](#)
- [CopyKeyColorName](#)
- [CopyKeysColor](#)
- [CopyKeysColorAllFrames](#)
- [CopyKeysColorAllFramesName](#)
- [CopyKeysColorName](#)
- [CopyNonZeroAllKeys](#)
- [CopyNonZeroAllKeysAllFrames](#)
- [CopyNonZeroAllKeysAllFramesName](#)
- [CopyNonZeroAllKeysAllFramesOffset](#)
- [CopyNonZeroAllKeysAllFramesOffsetName](#)
- [CopyNonZeroAllKeysName](#)
- [CopyNonZeroAllKeysOffset](#)
- [CopyNonZeroAllKeysOffsetName](#)
- [CopyNonZeroKeyColor](#)
- [CopyNonZeroKeyColorName](#)

- [CopyNonZeroTargetAllKeys](#)
- [CopyNonZeroTargetAllKeysAllFrames](#)
- [CopyNonZeroTargetAllKeysAllFramesName](#)
- [CopyNonZeroTargetAllKeysAllFramesOffset](#)
- [CopyNonZeroTargetAllKeysAllFramesOffsetName](#)
- [CopyNonZeroTargetAllKeysName](#)
- [CopyZeroTargetAllKeysAllFrames](#)
- [CopyZeroTargetAllKeysAllFramesName](#)
- [DuplicateFirstFrame](#)
- [DuplicateFirstFrameName](#)
- [DuplicateFrames](#)
- [DuplicateFramesName](#)
- [DuplicateMirrorFrames](#)
- [DuplicateMirrorFramesName](#)
- [FadeEndFrames](#)
- [FadeEndFramesName](#)
- [FadeStartFrames](#)
- [FadeStartFramesName](#)
- [FillColor](#)
- [FillColorAllFrames](#)
- [FillColorAllFramesName](#)
- [FillColorAllFramesRGB](#)
- [FillColorAllFramesRGBName](#)
- [FillColorName](#)
- [FillColorRGB](#)
- [FillColorRGBName](#)
- [FillNonZeroColor](#)
- [FillNonZeroColorAllFrames](#)

- [FillNonZeroColorAllFramesName](#)
- [FillNonZeroColorAllFramesRGB](#)
- [FillNonZeroColorAllFramesRGBName](#)
- [FillNonZeroColorName](#)
- [FillNonZeroColorRGB](#)
- [FillNonZeroColorRGBName](#)
- [FillRandomColors](#)
- [FillRandomColorsAllFrames](#)
- [FillRandomColorsAllFramesName](#)
- [FillRandomColorsBlackAndWhite](#)
- [FillRandomColorsBlackAndWhiteAllFrames](#)
- [FillRandomColorsBlackAndWhiteAllFramesName](#)
- [FillRandomColorsBlackAndWhiteName](#)
- [FillRandomColorsName](#)
- [FillThresholdColorsAllFrames](#)
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- [FillThresholdColorsRGBName](#)
- [FillThresholdRGBColorsAllFramesRGB](#)
- [FillThresholdRGBColorsAllFramesRGBName](#)
- [FillZeroColor](#)
- [FillZeroColorAllFrames](#)
- [FillZeroColorAllFramesName](#)
- [FillZeroColorAllFramesRGB](#)

- [FillZeroColorAllFramesRGBName](#)
- [FillZeroColorName](#)
- [FillZeroColorRGB](#)
- [FillZeroColorRGBName](#)
- [GetAnimation](#)
- [GetAnimationCount](#)
- [GetAnimationId](#)
- [GetAnimationName](#)
- [GetCurrentFrame](#)
- [GetCurrentFrameName](#)
- [GetFrameCount](#)
- [GetFrameCountName](#)
- [GetFrameDuration](#)
- [GetFrameDurationName](#)
- [GetKeyColor](#)
- [GetKeyColorName](#)
- [GetMaxColumn](#)
- [GetMaxLeds](#)
- [GetMaxRow](#)
- [GetPlayingAnimationCount](#)
- [GetPlayingAnimationId](#)
- [GetRGB](#)
- [GetTotalDuration](#)
- [GetTotalDurationName](#)
- [InsertDelay](#)
- [InsertDelayName](#)
- [InsertFrame](#)
- [InsertFrameName](#)

- [InvertColorsAllFrames](#)
- [InvertColorsAllFramesName](#)
- [IsActive](#)
- [IsConnected](#)
- [IsInitialized](#)
- [Lerp](#)
- [LerpColor](#)
- [LoadAnimation](#)
- [LoadAnimationName](#)
- [MakeBlankFrames](#)
- [MakeBlankFramesName](#)
- [MakeBlankFramesRandom](#)
- [MakeBlankFramesRandomBlackAndWhite](#)
- [MakeBlankFramesRandomBlackAndWhiteName](#)
- [MakeBlankFramesRandomName](#)
- [MakeBlankFramesRGB](#)
- [MakeBlankFramesRGBName](#)
- [MultiplyColorLerpAllFrames](#)
- [MultiplyColorLerpAllFramesName](#)
- [MultiplyIntensity](#)
- [MultiplyIntensityAllFrames](#)
- [MultiplyIntensityAllFramesName](#)
- [MultiplyIntensityAllFramesRGB](#)
- [MultiplyIntensityAllFramesRGBName](#)
- [MultiplyIntensityColor](#)
- [MultiplyIntensityColorAllFrames](#)
- [MultiplyIntensityColorAllFramesName](#)
- [MultiplyIntensityColorName](#)

- [MultiplyIntensityName](#)
- [MultiplyIntensityRGB](#)
- [MultiplyIntensityRGBName](#)
- [MultiplyNonZeroTargetColorLerpAllFrames](#)
- [MultiplyNonZeroTargetColorLerpAllFramesName](#)
- [MultiplyTargetColorLerpAllFrames](#)
- [MultiplyTargetColorLerpAllFramesName](#)
- [OffsetColors](#)
- [OffsetColorsAllFrames](#)
- [OffsetColorsAllFramesName](#)
- [OffsetColorsName](#)
- [OffsetNonZeroColors](#)
- [OffsetNonZeroColorsAllFrames](#)
- [OffsetNonZeroColorsAllFramesName](#)
- [OffsetNonZeroColorsName](#)
- [OpenAnimationFromMemory](#)
- [OverrideFrameDurationName](#)
- [PlayAnimation](#)
- [PlayAnimationName](#)
- [PreviewFrame](#)
- [PreviewFrameName](#)
- [ReduceFrames](#)
- [ReduceFramesName](#)
- [ReverseAllFrames](#)
- [ReverseAllFramesName](#)
- [SetChromaCustomColorAllFramesName](#)
- [SetChromaCustomFlagName](#)
- [SetCurrentFrame](#)

- [SetCurrentFrameName](#)
- [SetEventName](#)
- [SetIdleAnimationName](#)
- [SetKeyColor](#)
- [SetKeyColorAllFrames](#)
- [SetKeyColorAllFramesName](#)
- [SetKeyColorName](#)
- [SetKeyNonZeroColor](#)
- [SetKeyNonZeroColorName](#)
- [SetKeyRowColumnColorName](#)
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- 

### AddNonZeroAllKeys

Add source color to target where color is not black for frame id, reference source and target by id.

```
void UChromaSDKPluginBPLibrary::AddNonZeroAllKeys(int32 sourceAnimationId,
int32 targetAnimationId, int32 frameId);
```

---

### AddNonZeroAllKeysAllFrames

Add source color to target where color is not black for all frames, reference source and target by id.

```
void UChromaSDKPluginBPLibrary::AddNonZeroAllKeysAllFrames(int32
sourceAnimationId,
int32 targetAnimationId);
```

---

### AddNonZeroAllKeysAllFramesName

Add source color to target where color is not black for all frames, reference source and target by name.

```
void UChromaSDKPluginBPLibrary::AddNonZeroAllKeysAllFramesName(const FString&
sourceAnimationName, const FString& targetAnimationName);
```

---

### AddNonZeroAllKeysAllFramesOffset

Add source color to target where color is not black for all frames starting at offset for the length of the source, reference source and target by id.

```
void UChromaSDKPluginBPLibrary::AddNonZeroAllKeysAllFramesOffset(int32
sourceAnimationId,
int32 targetAnimationId, int32 offset);
```

---

### AddNonZeroAllKeysAllFramesOffsetName

Add source color to target where color is not black for all frames starting at offset for the length of the source, reference source and target by name.

```
void UChromaSDKPluginBPLibrary::AddNonZeroAllKeysAllFramesOffsetName(const
    FString& sourceAnimationName, const FString& targetAnimationName, int32
    offset);
```

---

### AddNonZeroAllKeysName

Add source color to target where color is not black for frame id, reference source and target by name.

```
void UChromaSDKPluginBPLibrary::AddNonZeroAllKeysName(const FString&
    sourceAnimationName,
    const FString& targetAnimationName, int32 frameId);
```

---

### AddNonZeroTargetAllKeysAllFrames

Add source color to target where the target color is not black for all frames, reference source and target by id.

```
void UChromaSDKPluginBPLibrary::AddNonZeroTargetAllKeysAllFrames(int32
    sourceAnimationId,
    int32 targetAnimationId);
```

---

### AddNonZeroTargetAllKeysAllFramesName

Add source color to target where the target color is not black for all frames, reference source and target by name.

```
void UChromaSDKPluginBPLibrary::AddNonZeroTargetAllKeysAllFramesName(const
    FString& sourceAnimationName, const FString& targetAnimationName);
```

---

### AddNonZeroTargetAllKeysAllFramesOffset

Add source color to target where the target color is not black for all frames starting at offset for the length of the source, reference source and target by id.

```
void UChromaSDKPluginBPLibrary::AddNonZeroTargetAllKeysAllFramesOffset(int32
    sourceAnimationId, int32 targetAnimationId, int32 offset);
```

---

### AddNonZeroTargetAllKeysAllFramesOffsetName

Add source color to target where the target color is not black for all frames starting at offset for the length of the source, reference source and target by name.

```
void UChromaSDKPluginBPLibrary::AddNonZeroTargetAllKeysAllFramesOffsetName(const
    FString& sourceAnimationName, const FString& targetAnimationName, int32
    offset);
```

---

### AppendAllFrames

Append all source frames to the target animation, reference source and target by id.

```
void UChromaSDKPluginBPLibrary::AppendAllFrames(int32 sourceAnimationId,
    int32 targetAnimationId);
```

---

### AppendAllFramesName

Append all source frames to the target animation, reference source and target by name.

```
void UChromaSDKPluginBPLibrary::AppendAllFramesName(const FString&
    sourceAnimationName,
    const FString& targetAnimationName);
```

---

### ClearAll

`PluginClearAll` will issue a `CLEAR` effect for all devices.

```
void UChromaSDKPluginBPLibrary::ClearAll();
```

---

### ClearAnimationType

`PluginClearAnimationType` will issue a `CLEAR` effect for the given device.

```
void UChromaSDKPluginBPLibrary::ClearAnimationType(EChromaSDKDeviceEnum::Type
    device);
```

---

## CloseAll

`PluginCloseAll` closes all open animations so they can be reloaded from disk. The set of animations will be stopped if playing.

```
void UChromaSDKPluginBPLibrary::CloseAll();
```

---

## CloseAnimation

Closes the `Chroma` animation to free up resources referenced by id. Returns the animation id upon success. Returns negative one upon failure. This might be used while authoring effects if there was a change necessitating re-opening the animation. The animation id can no longer be used once closed.

```
void UChromaSDKPluginBPLibrary::CloseAnimation(const int32 animationId);
```

---

## CloseAnimationName

Closes the `Chroma` animation referenced by name so that the animation can be reloaded from disk.

```
void UChromaSDKPluginBPLibrary::CloseAnimationName(const FString& animationName);
```

---

## CopyAllKeys

Copy source animation to target animation for the given frame. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyAllKeys(int32 sourceAnimationId, int32  
targetAnimationId, int32 frameId);
```

---

## CopyAllKeysName

Copy source animation to target animation for the given frame. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyAllKeysName(const FString&  
sourceAnimationName,  
const FString& targetAnimationName, int32 frameId);
```

---

## CopyAnimation

Copy animation to named target animation in memory. If target animation exists, close first. Source is referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyAnimation(int32 sourceAnimationId, const
FString& targetAnimationName);
```

---

### CopyAnimationName

Copy animation to named target animation in memory. If target animation exists, close first. Source is referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyAnimationName(const FString&
sourceAnimationName,
const FString& targetAnimationName);
```

---

### CopyKeyColor

Copy animation key color from the source animation to the target animation for the given frame. Reference the source and target by id.

```
void UChromaSDKPluginBPLibrary::CopyKeyColor(int32 sourceAnimationId, int32
targetAnimationId, int32 frameIndex, EChromaSDKKeyboardKey::Type key);
```

---

### CopyKeyColorName

Copy animation key color from the source animation to the target animation for the given frame.

```
void UChromaSDKPluginBPLibrary::CopyKeyColorName(const FString&
sourceAnimationName,
const FString& targetAnimationName, const int32 frameIndex,
EChromaSDKKeyboardKey::Type
key);
```

---

### CopyKeysColor

Copy animation color for a set of keys from the source animation to the target animation for the given frame. Reference the source and target by id.

```
void UChromaSDKPluginBPLibrary::CopyKeysColor(int32 sourceAnimationId, int32
targetAnimationId, int32 frameIndex, const
```

```
TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys);
```

---

### CopyKeysColorAllFrames

Copy animation color for a set of keys from the source animation to the target animation for all frames. Reference the source and target by id.

```
void UChromaSDKPluginBPLibrary::CopyKeysColorAllFrames(int32 sourceAnimationId,
    int32 targetAnimationId, const
    TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys);
```

---

### CopyKeysColorAllFramesName

Copy animation color for a set of keys from the source animation to the target animation for all frames. Reference the source and target by name.

```
void UChromaSDKPluginBPLibrary::CopyKeysColorAllFramesName(const FString&
    sourceAnimationName, const FString& targetAnimationName, const
    TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys);
```

---

### CopyKeysColorName

Copy animation color for a set of keys from the source animation to the target animation for the given frame. Reference the source and target by name.

```
void UChromaSDKPluginBPLibrary::CopyKeysColorName(const FString&
    sourceAnimationName,
    const FString& targetAnimationName, const int32 frameIndex, const
    TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys);
```

---

### CopyNonZeroAllKeys

Copy source animation to target animation for the given frame. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeys(int32 sourceAnimationId,
    int32 targetAnimationId, int32 frameId);
```

---

### CopyNonZeroAllKeysAllFrames

Copy nonzero colors from a source animation to a target animation for all frames. Reference source and target by id.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeysAllFrames(int32
sourceAnimationId,
int32 targetAnimationId);
```

---

### CopyNonZeroAllKeysAllFramesName

Copy nonzero colors from a source animation to a target animation for all frames. Reference source and target by name.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeysAllFramesName(const FString&
sourceAnimationName, const FString& targetAnimationName);
```

---

### CopyNonZeroAllKeysAllFramesOffset

Copy nonzero colors from a source animation to a target animation for all frames starting at the offset for the length of the source animation. The source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeysAllFramesOffset(int32
sourceAnimationId, int32 targetAnimationId, int32 offset);
```

---

### CopyNonZeroAllKeysAllFramesOffsetName

Copy nonzero colors from a source animation to a target animation for all frames starting at the offset for the length of the source animation. The source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeysAllFramesOffsetName(const
FString& sourceAnimationName, const FString& targetAnimationName, int32
offset);
```

---

### CopyNonZeroAllKeysName

Copy nonzero colors from source animation to target animation for the specified frame. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeysName(const FString&
sourceAnimationName,
    const FString& targetAnimationName, int32 frameId);
```

---

### CopyNonZeroAllKeysOffset

Copy nonzero colors from the source animation to the target animation from the source frame to the target offset frame. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeysOffset(int32 sourceAnimationId,
    int32 targetAnimationId, int32 frameId, int32 offset);
```

---

### CopyNonZeroAllKeysOffsetName

Copy nonzero colors from the source animation to the target animation from the source frame to the target offset frame. Source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroAllKeysOffsetName(const FString&
sourceAnimationName, const FString& targetAnimationName, int32 frameId,
    int32 offset);
```

---

### CopyNonZeroKeyColor

Copy animation key color from the source animation to the target animation for the given frame where color is not zero.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroKeyColor(int32 sourceAnimationId,
    int32 targetAnimationId, int32 frameIndex, EChromaSDKKeyboardKey::Type
    key);
```

---

### CopyNonZeroKeyColorName

Copy animation key color from the source animation to the target animation for the given frame where color is not zero.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroKeyColorName(const FString&
sourceAnimationName,
    const FString& targetAnimationName, const int32 frameIndex,
    EChromaSDKKeyboardKey::Type
    key);
```

---

### CopyNonZeroTargetAllKeys

Copy nonzero colors from the source animation to the target animation where the target color is nonzero for the specified frame. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroTargetAllKeys(int32 sourceAnimationId,
    int32 targetAnimationId, int32 frameId);
```

---

### CopyNonZeroTargetAllKeysAllFrames

Copy nonzero colors from the source animation to the target animation where the target color is nonzero for all frames. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroTargetAllKeysAllFrames(int32
    sourceAnimationId, int32 targetAnimationId);
```

---

### CopyNonZeroTargetAllKeysAllFramesName

Copy nonzero colors from the source animation to the target animation where the target color is nonzero for all frames. Source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroTargetAllKeysAllFramesName(const
    FString& sourceAnimationName, const FString& targetAnimationName);
```

---

### CopyNonZeroTargetAllKeysAllFramesOffset

Copy nonzero colors from the source animation to the target animation where the target color is nonzero for all frames. Source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroTargetAllKeysAllFramesOffset(int32
    sourceAnimationId, int32 targetAnimationId, int32 offset);
```

---

### CopyNonZeroTargetAllKeysAllFramesOffsetName

Copy nonzero colors from the source animation to the target animation where the target color is nonzero for all frames starting at the target offset for the length of the source animation. Source and target animations are referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroTargetAllKeysAllFramesOffsetName(const FString& sourceAnimationName, const FString& targetAnimationName, int32 offset);
```

---

### CopyNonZeroTargetAllKeysName

Copy nonzero colors from the source animation to the target animation where the target color is nonzero for the specified frame. The source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyNonZeroTargetAllKeysName(const FString& sourceAnimationName, const FString& targetAnimationName, int32 frameId);
```

---

### CopyZeroTargetAllKeysAllFrames

Copy nonzero color from source animation to target animation where target is zero for all frames. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::CopyZeroTargetAllKeysAllFrames(int32 sourceAnimationId, int32 targetAnimationId);
```

---

### CopyZeroTargetAllKeysAllFramesName

Copy nonzero color from source animation to target animation where target is zero for all frames. Source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::CopyZeroTargetAllKeysAllFramesName(const FString& sourceAnimationName, const FString& targetAnimationName);
```

---

### DuplicateFirstFrame

Duplicate the first animation frame so that the animation length matches the frame count. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::DuplicateFirstFrame(int32 animationId, int32 frameCount);
```

---

### DuplicateFirstFrameName

Duplicate the first animation frame so that the animation length matches the frame count. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::DuplicateFirstFrameName(const FString&
    animationName,
    int32 frameCount);
```

---

### DuplicateFrames

Duplicate all the frames of the animation to double the animation length. Frame 1 becomes frame 1 and 2. Frame 2 becomes frame 3 and 4. And so on. The animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::DuplicateFrames(int32 animationId);
```

---

### DuplicateFramesName

Duplicate all the frames of the animation to double the animation length. Frame 1 becomes frame 1 and 2. Frame 2 becomes frame 3 and 4. And so on. The animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::DuplicateFramesName(const FString& animationName);
```

---

### DuplicateMirrorFrames

Duplicate all the animation frames in reverse so that the animation plays forwards and backwards. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::DuplicateMirrorFrames(int32 animationId);
```

---

### DuplicateMirrorFramesName

Duplicate all the animation frames in reverse so that the animation plays forwards and backwards. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::DuplicateMirrorFramesName(const FString&
    animationName);
```

---

### FadeEndFrames

Fade the animation to black starting at the fade frame index to the end of the animation. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FadeEndFrames(int32 animationId, int32 fade);
```

---

### FadeEndFramesName

Fade the animation to black starting at the fade frame index to the end of the animation. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FadeEndFramesName(const FString& animationName,  
int32 fade);
```

---

### FadeStartFrames

Fade the animation from black to full color starting at 0 to the fade frame index. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FadeStartFrames(int32 animationId, int32  
fade);
```

---

### FadeStartFramesName

Fade the animation from black to full color starting at 0 to the fade frame index. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FadeStartFramesName(const FString& animationName,  
int32 fade);
```

---

### FillColor

Set the RGB value for all colors in the specified frame. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillColor(int32 animationId, int32 frameId,  
const FLinearColor& colorParam);
```

---

### FillColorAllFrames

Set the RGB value for all colors for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillColorAllFrames(int32 animationId, const
    FLinearColor& colorParam);
```

---

### FillColorAllFramesName

Set the RGB value for all colors for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillColorAllFramesName(const FString&
    animationName,
    const FLinearColor& colorParam);
```

---

### FillColorAllFramesRGB

Set the RGB value for all colors for all frames. Use the range of 0 to 255 for red, green, and blue parameters. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillColorAllFramesRGB(int32 animationId,
    int32 red, int32 green, int32 blue);
```

---

### FillColorAllFramesRGBName

Set the RGB value for all colors for all frames. Use the range of 0 to 255 for red, green, and blue parameters. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillColorAllFramesRGBName(const FString&
    animationName, int32 red, int32 green, int32 blue);
```

---

### FillColorName

Set the RGB value for all colors in the specified frame. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillColorName(const FString& animationName,
    int32 frameId, const FLinearColor& colorParam);
```

---

### FillColorRGB

Set the RGB value for all colors in the specified frame. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillColorRGB(int32 animationId, int32 frameId,
int32 red, int32 green, int32 blue);
```

---

### FillColorRGBName

Set the RGB value for all colors in the specified frame. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillColorRGBName(const FString& animationName,
int32 frameId, int32 red, int32 green, int32 blue);
```

---

### FillNonZeroColor

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors in the specified frame. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColor(int32 animationId, int32
frameId, const FLinearColor& colorParam);
```

---

### FillNonZeroColorAllFrames

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColorAllFrames(int32 animationId,
const FLinearColor& colorParam);
```

---

### FillNonZeroColorAllFramesName

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColorAllFramesName(const FString&
animationName, const FLinearColor& colorParam);
```

---

### FillNonZeroColorAllFramesRGB

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors for all frames. Use the range of 0 to 255 for red, green, and blue parameters. Animation is

referenced by id.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColorAllFramesRGB(int32 animationId,
    int32 red, int32 green, int32 blue);
```

---

### FillNonZeroColorAllFramesRGBName

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors for all frames. Use the range of 0 to 255 for red, green, and blue parameters. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColorAllFramesRGBName(const FString&
    animationName, int32 red, int32 green, int32 blue);
```

---

### FillNonZeroColorName

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors in the specified frame. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColorName(const FString& animationName,
    int32 frameId, const FLinearColor& colorParam);
```

---

### FillNonZeroColorRGB

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors in the specified frame. Use the range of 0 to 255 for red, green, and blue parameters. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColorRGB(int32 animationId, int32
    frameId, int32 red, int32 green, int32 blue);
```

---

### FillNonZeroColorRGBName

This method will only update colors in the animation that are not already set to black. Set the RGB value for a subset of colors in the specified frame. Use the range of 0 to 255 for red, green, and blue parameters. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillNonZeroColorRGBName(const FString&
    animationName,
```

```
int32 frameId, int32 red, int32 green, int32 blue);
```

---

### FillRandomColors

Fill the frame with random RGB values for the given frame. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillRandomColors(int32 animationId, int32  
frameId);
```

---

### FillRandomColorsAllFrames

Fill the frame with random RGB values for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillRandomColorsAllFrames(int32 animationId);
```

---

### FillRandomColorsAllFramesName

Fill the frame with random RGB values for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillRandomColorsAllFramesName(const FString&  
animationName);
```

---

### FillRandomColorsBlackAndWhite

Fill the frame with random black and white values for the specified frame. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillRandomColorsBlackAndWhite(int32 animationId,  
int32 frameId);
```

---

### FillRandomColorsBlackAndWhiteAllFrames

Fill the frame with random black and white values for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillRandomColorsBlackAndWhiteAllFrames(int32  
animationId);
```

---

### FillRandomColorsBlackAndWhiteAllFramesName

Fill the frame with random black and white values for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillRandomColorsBlackAndWhiteAllFramesName(const FString& animationName);
```

---

### FillRandomColorsBlackAndWhiteName

Fill the frame with random black and white values for the specified frame. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillRandomColorsBlackAndWhiteName(const FString& animationName, int32 frameId);
```

---

### FillRandomColorsName

Fill the frame with random RGB values for the given frame. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillRandomColorsName(const FString& animationName, int32 frameId);
```

---

### FillThresholdColorsAllFrames

Fill all frames with RGB color where the animation color is less than the RGB threshold. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsAllFrames(int32 animationId, int32 threshold, const FLinearColor& colorParam);
```

---

### FillThresholdColorsAllFramesName

Fill all frames with RGB color where the animation color is less than the RGB threshold. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsAllFramesName(const FString& animationName, int32 threshold, const FLinearColor& colorParam);
```

---

### FillThresholdColorsAllFramesRGB

Fill all frames with RGB color where the animation color is less than the threshold. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsAllFramesRGB(int32 animationId,
    int32 threshold, int32 red, int32 green, int32 blue);
```

---

### FillThresholdColorsAllFramesRGBName

Fill all frames with RGB color where the animation color is less than the threshold. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsAllFramesRGBName(const
    FString& animationName, int32 threshold, int32 red, int32 green, int32
    blue);
```

---

### FillThresholdColorsMinMaxAllFramesRGB

Fill all frames with the min RGB color where the animation color is less than the min threshold AND with the max RGB color where the animation is more than the max threshold. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsMinMaxAllFramesRGB(int32
    animationId, int32 minThreshold, int32 minRed, int32 minGreen, int32 minBlue,
    int32 maxThreshold, int32 maxRed, int32 maxGreen, int32 maxBlue);
```

---

### FillThresholdColorsMinMaxAllFramesRGBName

Fill all frames with the min RGB color where the animation color is less than the min threshold AND with the max RGB color where the animation is more than the max threshold. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsMinMaxAllFramesRGBName(const
    FString& animationName, int32 minThreshold, int32 minRed, int32 minGreen,
    int32 minBlue, int32 maxThreshold, int32 maxRed, int32 maxGreen, int32
    maxBlue);
```

---

### FillThresholdColorsRGB

Fill the specified frame with RGB color where the animation color is less than the RGB threshold. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsRGB(int32 animationId,
int32 frameId, int32 threshold, int32 red, int32 green, int32 blue);
```

---

### FillThresholdColorsRGBName

Fill the specified frame with RGB color where the animation color is less than the RGB threshold. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillThresholdColorsRGBName(const FString&
animationName, int32 frameId, int32 threshold, int32 red, int32 green,
int32 blue);
```

---

### FillThresholdRGBColorsAllFramesRGB

Fill all frames with RGB color where the animation color is less than the RGB threshold. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillThresholdRGBColorsAllFramesRGB(int32
animationId, int32 redThreshold, int32 greenThreshold, int32 blueThreshold,
int32 red, int32 green, int32 blue);
```

---

### FillThresholdRGBColorsAllFramesRGBName

Fill all frames with RGB color where the animation color is less than the RGB threshold. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillThresholdRGBColorsAllFramesRGBName(const
FString& animationName, int32 redThreshold, int32 greenThreshold, int32
blueThreshold, int32 red, int32 green, int32 blue);
```

---

### FillZeroColor

Fill the specified frame with RGB color where the animation color is zero. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillZeroColor(int32 animationId, int32 frameId,
const FLinearColor& colorParam);
```

---

### FillZeroColorAllFrames

Fill all frames with RGB color where the animation color is zero. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillZeroColorAllFrames(int32 animationId,  
    const FLinearColor& colorParam);
```

---

### FillZeroColorAllFramesName

Fill all frames with RGB color where the animation color is zero. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillZeroColorAllFramesName(const FString&  
    animationName, const FLinearColor& colorParam);
```

---

### FillZeroColorAllFramesRGB

Fill all frames with RGB color where the animation color is zero. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillZeroColorAllFramesRGB(int32 animationId,  
    int32 red, int32 green, int32 blue);
```

---

### FillZeroColorAllFramesRGBName

Fill all frames with RGB color where the animation color is zero. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillZeroColorAllFramesRGBName(const FString&  
    animationName, int32 red, int32 green, int32 blue);
```

---

### FillZeroColorName

Fill the specified frame with RGB color where the animation color is zero. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillZeroColorName(const FString& animationName,  
    int32 frameId, const FLinearColor& colorParam);
```

---

### FillZeroColorRGB

Fill the specified frame with RGB color where the animation color is zero. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::FillZeroColorRGB(int32 animationId, int32
    frameId, int32 red, int32 green, int32 blue);
```

---

### FillZeroColorRGBName

Fill the specified frame with RGB color where the animation color is zero. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::FillZeroColorRGBName(const FString& animationName,
    int32 frameId, int32 red, int32 green, int32 blue);
```

---

### GetAnimation

Get the animation id for the named animation.

```
int32 UChromaSDKPluginBPLibrary::GetAnimation(const FString& animationName);
```

---

### GetAnimationCount

`PluginGetAnimationCount` will return the number of loaded animations.

```
int32 UChromaSDKPluginBPLibrary::GetAnimationCount();
```

---

### GetAnimationId

`PluginGetAnimationId` will return the `animationId` given the `index` of the loaded animation. The `index` is zero-based and less than the number returned by `PluginGetAnimationCount`. Use `PluginGetAnimationName` to get the name of the animation.

```
int32 UChromaSDKPluginBPLibrary::GetAnimationId(const FString& animationName);
```

---

### GetAnimationName

`PluginGetAnimationName` takes an `animationId` and returns the name of the animation of the `.chroma` animation file. If a name is not available then an empty string will be returned.

```
FString UChromaSDKPluginBPLibrary::GetAnimationName(const int32 animationId);
```

---

### GetCurrentFrame

Get the current frame of the animation referenced by id.

```
int32 UChromaSDKPluginBPLibrary::GetCurrentFrame(int32 animationId);
```

---

### GetCurrentFrameName

Get the current frame of the animation referenced by name.

```
int32 UChromaSDKPluginBPLibrary::GetCurrentFrameName(const FString& animationName);
```

---

### GetFrameCount

Returns the frame count of a **Chroma** animation upon success. Returns negative one upon failure.

```
int32 UChromaSDKPluginBPLibrary::GetFrameCount(const int32 animationId);
```

---

### GetFrameCountName

Returns the frame count of a **Chroma** animation upon success. Returns negative one upon failure.

```
int32 UChromaSDKPluginBPLibrary::GetFrameCountName(const FString& animationName);
```

---

### GetFrameDuration

Returns the duration of an animation frame in seconds upon success. Returns zero upon failure.

```
float UChromaSDKPluginBPLibrary::GetFrameDuration(int32 animationId, int32 frameId);
```

---

### GetFrameDurationName

Returns the duration of an animation frame in seconds upon success. Returns zero upon failure.

```
float UChromaSDKPluginBPLibrary::GetFrameDurationName(const FString&
    animationName,
    int32 frameId);
```

---

### GetKeyColor

Get the color of an animation key for the given frame referenced by id.

```
FLinearColor UChromaSDKPluginBPLibrary::GetKeyColor(int32 animationId, int32
    frameIndex, EChromaSDKKeyboardKey::Type key);
```

---

### GetKeyColorName

Get the color of an animation key for the given frame referenced by name.

```
FLinearColor UChromaSDKPluginBPLibrary::GetKeyColorName(const FString&
    animationName,
    const int32 frameIndex, EChromaSDKKeyboardKey::Type key);
```

---

### GetMaxColumn

Returns the MAX COLUMN given the EChromaSDKDevice2DEnum device as an integer upon success. Returns negative one upon failure.

```
int32 UChromaSDKPluginBPLibrary::GetMaxColumn(EChromaSDKDevice2DEnum::Type
    device);
```

---

### GetMaxLeds

Returns the MAX LEDS given the EChromaSDKDevice1DEnum device as an integer upon success. Returns negative one upon failure.

```
int32 UChromaSDKPluginBPLibrary::GetMaxLeds(EChromaSDKDevice1DEnum::Type
    device);
```

---

### GetMaxRow

Returns the **MAX ROW** given the **EChromaSDKDevice2DEnum** device as an integer upon success. Returns negative one upon failure.

```
int32 UChromaSDKPluginBPLibrary::GetMaxRow(EChromaSDKDevice2DEnum::Type
device);
```

---

### GetPlayingAnimationCount

**PluginGetPlayingAnimationCount** will return the number of playing animations.

```
int32 UChromaSDKPluginBPLibrary::GetPlayingAnimationCount();
```

---

### GetPlayingAnimationId

**PluginGetPlayingAnimationId** will return the **animationId** given the **index** of the playing animation. The **index** is zero-based and less than the number returned by **PluginGetPlayingAnimationCount**. Use **PluginGetAnimationName** to get the name of the animation.

```
int32 UChromaSDKPluginBPLibrary::GetPlayingAnimationId(int32 index);
```

---

### GetRGB

Get the RGB color given red, green, and blue.

```
FLinearColor UChromaSDKPluginBPLibrary::GetRGB(int32 red, int32 green, int32
blue);
```

---

### GetTotalDuration

Returns the total duration of an animation in seconds upon success. Returns zero upon failure.

```
float UChromaSDKPluginBPLibrary::GetTotalDuration(int32 animationId);
```

---

### GetTotalDurationName

Returns the total duration of an animation in seconds upon success. Returns zero upon failure.

```
float UChromaSDKPluginBPLibrary::GetTotalDurationName(const FString&
animationName);
```

---

## InsertDelay

Insert an animation delay by duplicating the frame by the delay number of times. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::InsertDelay(int32 animationId, int32 frameId,
int32 delay);
```

---

## InsertDelayName

Insert an animation delay by duplicating the frame by the delay number of times. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::InsertDelayName(const FString& animationName,
int32 frameId, int32 delay);
```

---

## InsertFrame

Duplicate the source frame index at the target frame index. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::InsertFrame(int32 animationId, int32 sourceFrame,
int32 targetFrame);
```

---

## InsertFrameName

Duplicate the source frame index at the target frame index. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::InsertFrameName(const FString& animationName,
int32 sourceFrame, int32 targetFrame);
```

---

## InvertColorsAllFrames

Invert all the colors for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::InvertColorsAllFrames(int32 animationId);
```

---

## InvertColorsAllFramesName

Invert all the colors for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::InvertColorsAllFramesName(const FString&
    animationName);
```

---

## IsActive

Direct access to low level API.

```
int32 UChromaSDKPluginBPLibrary::IsActive(UPARAM(ref) bool& active);
```

---

## IsConnected

Direct access to low level API.

```
int32 UChromaSDKPluginBPLibrary::IsConnected(UPARAM(ref) FChromaSDKDeviceInfoType&
    deviceInfoType);
```

---

## IsInitialized

Returns true if the plugin has been initialized. Returns false if the plugin is uninitialized.

```
bool UChromaSDKPluginBPLibrary::IsInitialized();
```

---

## Lerp

Do a lerp math operation on a float.

```
float UChromaSDKPluginBPLibrary::Lerp(float start, float end, float amt);
```

---

## LerpColor

Lerp from one color to another given t in the range 0.0 to 1.0.

```
FLinearColor UChromaSDKPluginBPLibrary::LerpColor(FLinearColor colorParam1,  
    FLinearColor colorParam2, float t);
```

---

## LoadAnimation

Loads **Chroma** effects so that the animation can be played immediately. Returns the animation id upon success. Returns negative one upon failure.

```
void UChromaSDKPluginBPLibrary::LoadAnimation(const int32 animationId);
```

---

## LoadAnimationName

Load the named animation.

```
void UChromaSDKPluginBPLibrary::LoadAnimationName(const FString& animationName);
```

---

## MakeBlankFrames

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color defaults to color. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MakeBlankFrames(int32 animationId, int32  
    frameCount, float duration, const FLinearColor& colorParam);
```

---

## MakeBlankFramesName

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color defaults to color. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MakeBlankFramesName(const FString& animationName,  
    int32 frameCount, float duration, const FLinearColor& colorParam);
```

---

## MakeBlankFramesRandom

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color is random. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MakeBlankFramesRandom(int32 animationId,
int32 frameCount, float duration);
```

---

### MakeBlankFramesRandomBlackAndWhite

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color is random black and white. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MakeBlankFramesRandomBlackAndWhite(int32
animationId, int32 frameCount, float duration);
```

---

### MakeBlankFramesRandomBlackAndWhiteName

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color is random black and white. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MakeBlankFramesRandomBlackAndWhiteName(const
FString& animationName, int32 frameCount, float duration);
```

---

### MakeBlankFramesRandomName

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color is random. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MakeBlankFramesRandomName(const FString&
animationName, int32 frameCount, float duration);
```

---

### MakeBlankFramesRGB

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color defaults to color. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MakeBlankFramesRGB(int32 animationId, int32
frameCount, float duration, int32 red, int32 green, int32 blue);
```

---

### MakeBlankFramesRGBName

Make a blank animation for the length of the frame count. Frame duration defaults to the duration. The frame color defaults to color. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MakeBlankFramesRGBName(const FString&
    animationName,
    int32 frameCount, float duration, int32 red, int32 green, int32 blue);
```

---

### MultiplyColorLerpAllFrames

Multiply the color intensity with the lerp result from color 1 to color 2 using the frame index divided by the frame count for the `t` parameter. Animation is referenced in id.

```
void UChromaSDKPluginBPLibrary::MultiplyColorLerpAllFrames(int32 animationId,
    const FLinearColor& colorParam1, const FLinearColor& colorParam2);
```

---

### MultiplyColorLerpAllFramesName

Multiply the color intensity with the lerp result from color 1 to color 2 using the frame index divided by the frame count for the `t` parameter. Animation is referenced in name.

```
void UChromaSDKPluginBPLibrary::MultiplyColorLerpAllFramesName(const FString&
    animationName, const FLinearColor& colorParam1, const FLinearColor&
    colorParam2);
```

---

### MultiplyIntensity

Multiply all the colors in the frame by the intensity value. The valid the intensity range is from 0.0 to 255.0. RGB components are multiplied equally. An intensity of 0.5 would half the color value. Black colors in the frame will not be affected by this method.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensity(int32 animationId, int32
    frameId, float intensity);
```

---

### MultiplyIntensityAllFrames

Multiply all the colors for all frames by the intensity value. The valid the intensity range is from 0.0 to 255.0. RGB components are multiplied equally. An intensity of 0.5 would half the color value. Black colors in the frame will not be affected by this method.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityAllFrames(int32 animationId,
    float intensity);
```

---

### MultiplyIntensityAllFramesName

Multiply all the colors for all frames by the intensity value. The valid the intensity range is from 0.0 to 255.0. RGB components are multiplied equally. An intensity of 0.5 would half the color value. Black colors in the frame will not be affected by this method.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityAllFramesName(const FString&
    animationName, float intensity);
```

---

### MultiplyIntensityAllFramesRGB

Multiply all frames by the RBG color intensity. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityAllFramesRGB(int32 animationId,
    int32 red, int32 green, int32 blue);
```

---

### MultiplyIntensityAllFramesRGBName

Multiply all frames by the RBG color intensity. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityAllFramesRGBName(const
    FString& animationName, int32 red, int32 green, int32 blue);
```

---

### MultiplyIntensityColor

Multiply the specific frame by the RBG color intensity. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityColor(int32 animationId,
    int32 frameId, const FLinearColor& colorParam);
```

---

### MultiplyIntensityColorAllFrames

Multiply all frames by the RBG color intensity. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityColorAllFrames(int32 animationId,
    const FLinearColor& colorParam);
```

---

### MultiplyIntensityColorAllFramesName

Multiply all frames by the RGB color intensity. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityColorAllFramesName(const
    FString& animationName, const FLinearColor& colorParam);
```

---

### MultiplyIntensityColorName

Multiply the specific frame by the RGB color intensity. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityColorName(const FString&
    animationName, int32 frameId, const FLinearColor& colorParam);
```

---

### MultiplyIntensityName

Multiply all the colors in the frame by the intensity value. The valid the intensity range is from 0.0 to 255.0. RGB components are multiplied equally. An intensity of 0.5 would half the color value. Black colors in the frame will not be affected by this method.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityName(const FString&
    animationName,
    int32 frameId, float intensity);
```

---

### MultiplyIntensityRGB

Multiply the specific frame by the RGB color intensity. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityRGB(int32 animationId,
    int32 frameId, int32 red, int32 green, int32 blue);
```

---

### MultiplyIntensityRGBName

Multiply the specific frame by the RGB color intensity. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MultiplyIntensityRGBName(const FString&
    animationName, int32 frameId, int32 red, int32 green, int32 blue);
```

---

### MultiplyNonZeroTargetColorLerpAllFrames

Multiply all frames by the color lerp result between color 1 and 2 using the frame color value as the **t** value. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MultiplyNonZeroTargetColorLerpAllFrames(int32
    animationId, const FLinearColor& colorParam1, const FLinearColor&
    colorParam2);
```

---

### MultiplyNonZeroTargetColorLerpAllFramesName

Multiply all frames by the color lerp result between color 1 and 2 using the frame color value as the **t** value. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MultiplyNonZeroTargetColorLerpAllFramesName(const
    FString& animationName, const FLinearColor& colorParam1, const FLinearColor&
    colorParam2);
```

---

### MultiplyTargetColorLerpAllFrames

Multiply all frames by the color lerp result between color 1 and 2 using the frame color value as the **t** value. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::MultiplyTargetColorLerpAllFrames(int32
    animationId,
    const FLinearColor& colorParam1, const FLinearColor& colorParam2);
```

---

### MultiplyTargetColorLerpAllFramesName

Multiply all frames by the color lerp result between color 1 and 2 using the frame color value as the **t** value. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::MultiplyTargetColorLerpAllFramesName(const
    FString& animationName, const FLinearColor& colorParam1, const FLinearColor&
    colorParam2);
```

## OffsetColors

Offset all colors in the frame using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetColors(int32 animationId, int32 frameId,
                                             int32 red, int32 green, int32 blue);
```

---

## OffsetColorsAllFrames

Offset all colors for all frames using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetColorsAllFrames(int32 animationId,
                                                       int32 red, int32 green, int32 blue);
```

---

## OffsetColorsAllFramesName

Offset all colors for all frames using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetColorsAllFramesName(const FString&
                                                           animationName, int32 red, int32 green, int32 blue);
```

---

## OffsetColorsName

Offset all colors in the frame using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetColorsName(const FString& animationName,
                                                  int32 frameId, int32 red, int32 green, int32 blue);
```

---

## OffsetNonZeroColors

This method will only update colors in the animation that are not already set to black. Offset a subset of colors in the frame using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetNonZeroColors(int32 animationId, int32
                                                     frameId, int32 red, int32 green, int32 blue);
```

---

### OffsetNonZeroColorsAllFrames

This method will only update colors in the animation that are not already set to black. Offset a subset of colors for all frames using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetNonZeroColorsAllFrames(int32 animationId,
    int32 red, int32 green, int32 blue);
```

---

### OffsetNonZeroColorsAllFramesName

This method will only update colors in the animation that are not already set to black. Offset a subset of colors for all frames using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetNonZeroColorsAllFramesName(const FString&
    animationName, int32 red, int32 green, int32 blue);
```

---

### OffsetNonZeroColorsName

This method will only update colors in the animation that are not already set to black. Offset a subset of colors in the frame using the RGB offset. Use the range of -255 to 255 for red, green, and blue parameters. Negative values remove color. Positive values add color.

```
void UChromaSDKPluginBPLibrary::OffsetNonZeroColorsName(const FString&
    animationName,
    int32 frameId, int32 red, int32 green, int32 blue);
```

---

### OpenAnimationFromMemory

Opens a **Chroma** animation data from memory so that it can be played. **Data** is a pointer to BYTE array of the loaded animation in memory. **Name** will be assigned to the animation when loaded. Returns an animation id  $\geq 0$  upon success. Returns negative one if there was a failure. The animation id is used in most of the API methods.

```
void UChromaSDKPluginBPLibrary::OpenAnimationFromMemory(const TArray<uint8>&
    data, const FString& animationName);
```

## OverrideFrameDurationName

Override the duration of all frames with the `duration` value. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::OverrideFrameDurationName(const FString&
    animationName, float duration);
```

---

## PlayAnimation

Plays the `Chroma` animation. This will load the animation, if not loaded previously. Returns the animation id upon success. Returns negative one upon failure.

```
void UChromaSDKPluginBPLibrary::PlayAnimation(const FString& animationName,
    bool loop);
```

---

## PlayAnimationName

`PluginPlayAnimationName` automatically handles initializing the `ChromaSDK`. The named `.chroma` animation file will be automatically opened. The animation will play with looping `on` or `off`.

```
void UChromaSDKPluginBPLibrary::PlayAnimationName(const FString& animationName,
    bool loop);
```

---

## PreviewFrame

Displays the `Chroma` animation frame on `Chroma` hardware given the `frameId`. Returns the animation id upon success. Returns negative one upon failure.

```
int32 UChromaSDKPluginBPLibrary::PreviewFrame(int32 animationId, int32 frameId);
```

---

## PreviewFrameName

Displays the `Chroma` animation frame on `Chroma` hardware given the `frameId`. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::PreviewFrameName(const FString& animationName,
    int32 frameId);
```

---

## ReduceFrames

Reduce the frames of the animation by removing every nth element. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::ReduceFrames(int32 animationId, int32 n);
```

---

## ReduceFramesName

Reduce the frames of the animation by removing every nth element. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::ReduceFramesName(const FString& animationName, int32 n);
```

---

## ReverseAllFrames

Reverse the animation frame order of the **Chroma** animation. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::ReverseAllFrames(int32 animationId);
```

---

## ReverseAllFramesName

Reverse the animation frame order of the **Chroma** animation. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::ReverseAllFramesName(const FString& animationName);
```

---

## SetChromaCustomColorAllFramesName

When custom color is set, the custom key mode will be used. The animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::SetChromaCustomColorAllFramesName(const FString& animationName);
```

---

## SetChromaCustomFlagName

Set the Chroma custom key color flag on all frames. **True** changes the layout from grid to key. **True** changes the layout from key to grid. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::SetChromaCustomFlagName(const FString&
    animationName,
    bool flag);
```

---

### SetCurrentFrame

Set the current frame of the animation referenced by id.

```
void UChromaSDKPluginBPLibrary::SetCurrentFrame(int32 animationId, int32
    frameId);
```

---

### SetCurrentFrameName

Set the current frame of the animation referenced by name.

```
void UChromaSDKPluginBPLibrary::SetCurrentFrameName(const FString& animationName,
    int32 frameId);
```

---

### SetEventName

Direct access to low level API.

```
int32 UChromaSDKPluginBPLibrary::SetEventName(const FString& name);
```

---

### SetIdleAnimationName

When the idle animation is used, the named animation will play when no other animations are playing. Reference the animation by name.

```
void UChromaSDKPluginBPLibrary::SetIdleAnimationName(const FString&
    animationName);
```

---

### SetKeyColor

Set animation key to a static color for the given frame.

```
void UChromaSDKPluginBPLibrary::SetKeyColor(int32 animationId, int32 frameIndex,
    EChromaSDKKeyboardKey::Type key, const FLinearColor& colorParam);
```

---

### SetKeyColorAllFrames

Set the key to the specified key color for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::SetKeyColorAllFrames(int32 animationId,  
    EChromaSDKKeyboardKey::Type key, const FLinearColor& colorParam);
```

---

### SetKeyColorAllFramesName

Set the key to the specified key color for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::SetKeyColorAllFramesName(const FString&  
    animationName, EChromaSDKKeyboardKey::Type key, const FLinearColor&  
    colorParam);
```

---

### SetKeyColorName

Set animation key to a static color for the given frame.

```
void UChromaSDKPluginBPLibrary::SetKeyColorName(const FString& animationName,  
    const int32 frameIndex, EChromaSDKKeyboardKey::Type key, const FLinearColor&  
    colorParam);
```

---

### SetKeyNonZeroColor

Set animation key to a static color for the given frame if the existing color is not already black.

```
void UChromaSDKPluginBPLibrary::SetKeyNonZeroColor(int32 animationId, int32  
    frameIndex, EChromaSDKKeyboardKey::Type key, const FLinearColor& colorParam);
```

---

### SetKeyNonZeroColorName

Set animation key to a static color for the given frame if the existing color is not already black.

```
void UChromaSDKPluginBPLibrary::SetKeyNonZeroColorName(const FString&  
    animationName,  
    const int32 frameIndex, EChromaSDKKeyboardKey::Type key, const FLinearColor&  
    colorParam);
```

---

### SetKeyRowColumnColorName

Set animation key by row and column to a static color for the given frame.

```
void UChromaSDKPluginBPLibrary::SetKeyRowColumnColorName(const FString&
    animationName, const int32 frameIndex, const int32 row, const int32 column,
    const FLinearColor& colorParam);
```

---

### SetKeysColor

Set an array of animation keys to a static color for the given frame. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::SetKeysColor(int32 animationId, int32 frameIndex,
    const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>& keys, const
    FLinearColor&
    colorParam);
```

---

### SetKeysColorAllFrames

Set an array of animation keys to a static color for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::SetKeysColorAllFrames(int32 animationId,
    const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>& keys, const
    FLinearColor&
    colorParam);
```

---

### SetKeysColorAllFramesName

Set an array of animation keys to a static color for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::SetKeysColorAllFramesName(const FString&
    animationName, const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys, const FLinearColor& colorParam);
```

---

### SetKeysColorAllFramesRGB

Set an array of animation keys to a static color for all frames. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::SetKeysColorAllFramesRGB(int32 animationId,
    const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>& keys, int32 red,
    int32 green, int32 blue);
```

---

### SetKeysColorAllFramesRGBName

Set an array of animation keys to a static color for all frames. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::SetKeysColorAllFramesRGBName(const FString&
    animationName, const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys, int32 red, int32 green, int32 blue);
```

---

### SetKeysColorName

Set an array of animation keys to a static color for the given frame.

```
void UChromaSDKPluginBPLibrary::SetKeysColorName(const FString& animationName,
    const int32 frameIndex, const
    TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys, const FLinearColor& colorParam);
```

---

### SetKeysColorRGB

Set an array of animation keys to a static color for the given frame. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::SetKeysColorRGB(int32 animationId, int32
    frameIndex, const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>& keys,
    int32 red, int32 green, int32 blue);
```

---

### SetKeysColorRGBName

Set an array of animation keys to a static color for the given frame. Animation is referenced by name.

```
void UChromaSDKPluginBPLibrary::SetKeysColorRGBName(const FString& animationName,
    const int32 frameIndex, const
    TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys, int32 red, int32 green, int32 blue);
```

---

### SetKeysNonZeroColor

Set an array of animation keys to a static color for the given frame if the existing color is not already black.

```
void UChromaSDKPluginBPLibrary::SetKeysNonZeroColor(int32 animationId, int32
    frameIndex, const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>& keys,
    const FLinearColor& colorParam);
```

---

### SetKeysNonZeroColorAllFrames

Set an array of animation keys to a static color for the given frame where the color is not black. Animation is referenced by id.

```
void UChromaSDKPluginBPLibrary::SetKeysNonZeroColorAllFrames(int32 animationId,
    const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>& keys, const
    FLinearColor&
    colorParam);
```

---

### SetKeysNonZeroColorAllFramesName

Set an array of animation keys to a static color for all frames if the existing color is not already black. Reference animation by name.

```
void UChromaSDKPluginBPLibrary::SetKeysNonZeroColorAllFramesName(const FString&
    animationName, const TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys, const FLinearColor& colorParam);
```

---

### SetKeysNonZeroColorName

Set an array of animation keys to a static color for the given frame if the existing color is not already black. Reference animation by name.

```
void UChromaSDKPluginBPLibrary::SetKeysNonZeroColorName(const FString&
    animationName,
    const int32 frameIndex, const
    TArray<TEnumAsByte<EChromaSDKKeyboardKey::Type>>&
    keys, const FLinearColor& colorParam);
```

---

### SetStaticColor

Sets the target device to the static color.

```
void UChromaSDKPluginBPLibrary::SetStaticColor(EChromaSDKDeviceEnum::Type
device, const FLinearColor& color);
```

---

### SetStaticColorAll

Sets all devices to the static color.

```
void UChromaSDKPluginBPLibrary::SetStaticColorAll(const FLinearColor& color);
```

---

### StopAll

`PluginStopAll` will automatically stop all animations that are playing.

```
void UChromaSDKPluginBPLibrary::StopAll();
```

---

### StopAnimation

Stops animation playback if in progress. Returns the animation id upon success. Returns negative one upon failure.

```
void UChromaSDKPluginBPLibrary::StopAnimation(const FString& animationName);
```

---

### StopAnimationType

`PluginStopAnimationType` automatically handles initializing the `ChromaSDK`. If any animation is playing for the `deviceType` and `device` combination, it will be stopped.

```
void UChromaSDKPluginBPLibrary::StopAnimationType(EChromaSDKDeviceEnum::Type
device);
```

---

### StreamBroadcast

Begin broadcasting Chroma RGB data using the stored stream key as the endpoint. Intended for Cloud Gaming Platforms, restore the streaming key when the game instance is launched to continue streaming. `streamId` is a null terminated string `streamKey` is a null terminated string `StreamGetStatus()` should return the `READY` status to use this method.

```
void UChromaSDKPluginBPLibrary::StreamBroadcast(const FString& streamId,
const FString& streamKey);
```

---

### StreamBroadcastEnd

End broadcasting Chroma RGB data. StreamGetStatus() should return the BROADCASTING status to use this method.

```
void UChromaSDKPluginBPLibrary::StreamBroadcastEnd();
```

---

### StreamGetAuthShortcode

shortcode: Pass the address of a preallocated character buffer to get the streaming auth code. The buffer should have a minimum length of 6. length: Length will return as zero if the streaming auth code could not be obtained. If length is greater than zero, it will be the length of the returned streaming auth code. Once you have the shortcode, it should be shown to the user so they can associate the stream with their Razer ID StreamGetStatus() should return the READY status before invoking this method. platform: is the null terminated string that identifies the source of the stream: ( GEFORCE\_NOW, LUNA, STADIA, GAME\_PASS ) title: is the null terminated string that identifies the application or game.

```
FString UChromaSDKPluginBPLibrary::StreamGetAuthShortcode(const FString&
platform, const FString& title);
```

---

### StreamGetFocus

focus: Pass the address of a preallocated character buffer to get the stream focus. The buffer should have a length of 48 length: Length will return as zero if the stream focus could not be obtained. If length is greater than zero, it will be the length of the returned stream focus.

```
FString UChromaSDKPluginBPLibrary::StreamGetFocus();
```

---

### StreamGetId

Intended for Cloud Gaming Platforms, store the stream id to persist in user preferences to continue streaming if the game is suspended or closed. shortcode: The shortcode is a null terminated string. Use the shortcode that authorized the stream to obtain the stream id. streamId should be a preallocated buffer to get the stream key. The buffer should have a length of 48. length: Length will return zero if the key could not be obtained. If the length is greater than zero, it will be the length of the returned streaming id. Retrieve the stream id after

authorizing the shortcode. The authorization window will expire in 5 minutes. Be sure to save the stream key before the window expires. StreamGetStatus() should return the READY status to use this method.

```
FString UChromaSDKPluginBPLibrary::StreamGetId(const FString& shortcode);
```

---

### StreamGetKey

Intended for Cloud Gaming Platforms, store the streaming key to persist in user preferences to continue streaming if the game is suspended or closed. shortcode: The shortcode is a null terminated string. Use the shortcode that authorized the stream to obtain the stream key. If the status is in the BROADCASTING or WATCHING state, passing a NULL shortcode will return the active streamId. streamKey should be a preallocated buffer to get the stream key. The buffer should have a length of 48. length: Length will return zero if the key could not be obtained. If the length is greater than zero, it will be the length of the returned streaming key. Retrieve the stream key after authorizing the shortcode. The authorization window will expire in 5 minutes. Be sure to save the stream key before the window expires. StreamGetStatus() should return the READY status to use this method.

```
FString UChromaSDKPluginBPLibrary::StreamGetKey(const FString& shortcode);
```

---

### StreamGetStatusString

Convert StreamStatusType to a printable string

```
FString UChromaSDKPluginBPLibrary::StreamGetStatusString(const  
EChromaSDKStreamStatusEnum::Type  
status);
```

---

### StreamReleaseShortcode

This prevents the stream id and stream key from being obtained through the shortcode. This closes the auth window. shortcode is a null terminated string. StreamGetStatus() should return the READY status to use this method. returns success when shortcode has been released

```
bool UChromaSDKPluginBPLibrary::StreamReleaseShortcode(const FString& shortcode);
```

---

### StreamSetFocus

The focus is a null terminated string. Set the focus identifier for the application designated to automatically change the streaming state. Returns true on success.

```
bool UChromaSDKPluginBPLibrary::StreamSetFocus(const FString& streamFocus);
```

---

### StreamWatch

Begin watching the Chroma RGB data using streamID parameter. streamId is a null terminated string. StreamGetStatus() should return the READY status to use this method.

```
void UChromaSDKPluginBPLibrary::StreamWatch(const FString& streamId, int32  
    timestamp);
```

---

### StreamWatchEnd

End watching Chroma RGB data stream. StreamGetStatus() should return the WATCHING status to use this method.

```
void UChromaSDKPluginBPLibrary::StreamWatchEnd();
```

---

### SubtractNonZeroAllKeys

Subtract the source color from the target color for the frame where the target color is not black. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroAllKeys(int32 sourceAnimationId,  
    int32 targetAnimationId, int32 frameId);
```

---

### SubtractNonZeroAllKeysAllFrames

Subtract the source color from the target color for all frames where the target color is not black. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroAllKeysAllFrames(int32  
    sourceAnimationId,  
    int32 targetAnimationId);
```

---

### SubtractNonZeroAllKeysAllFramesName

Subtract the source color from the target color for all frames where the target color is not black. Source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroAllKeysAllFramesName(const
    FString& sourceAnimationName, const FString& targetAnimationName);
```

---

### SubtractNonZeroAllKeysAllFramesOffset

Subtract the source color from the target color for all frames where the target color is not black starting at offset for the length of the source. Source and target are referenced by id.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroAllKeysAllFramesOffset(int32
    sourceAnimationId, int32 targetAnimationId, int32 offset);
```

---

### SubtractNonZeroAllKeysAllFramesOffsetName

Subtract the source color from the target color for all frames where the target color is not black starting at offset for the length of the source. Source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroAllKeysAllFramesOffsetName(const
    FString& sourceAnimationName, const FString& targetAnimationName, int32
    offset);
```

---

### SubtractNonZeroAllKeysName

Subtract the source color from the target color for the frame where the target color is not black. Source and target are referenced by name.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroAllKeysName(const FString&
    sourceAnimationName, const FString& targetAnimationName, int32 frameId);
```

---

### SubtractNonZeroTargetAllKeysAllFrames

Subtract the source color from the target color where the target color is not black for all frames. Reference source and target by id.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroTargetAllKeysAllFrames(int32
    sourceAnimationId, int32 targetAnimationId);
```

---

### SubtractNonZeroTargetAllKeysAllFramesName

Subtract the source color from the target color where the target color is not black for all frames. Reference source and target by name.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroTargetAllKeysAllFramesName(const
    FString& sourceAnimationName, const FString& targetAnimationName);
```

---

### SubtractNonZeroTargetAllKeysAllFramesOffset

Subtract the source color from the target color where the target color is not black for all frames starting at the target offset for the length of the source. Reference source and target by id.

```
void UChromaSDKPluginBPLibrary::SubtractNonZeroTargetAllKeysAllFramesOffset(int32
    sourceAnimationId, int32 targetAnimationId, int32 offset);
```

---

### SubtractNonZeroTargetAllKeysAllFramesOffsetName

Subtract the source color from the target color where the target color is not black for all frames starting at the target offset for the length of the source. Reference source and target by name.

```
void
UChromaSDKPluginBPLibrary::SubtractNonZeroTargetAllKeysAllFramesOffsetName(const
    FString& sourceAnimationName, const FString& targetAnimationName, int32
    offset);
```

---

### TrimEndFrames

Trim the end of the animation. The length of the animation will be the lastFrameId plus one. Reference the animation by id.

```
void UChromaSDKPluginBPLibrary::TrimEndFrames(int32 animationId, int32
    lastFrameId);
```

---

### TrimEndFramesName

Trim the end of the animation. The length of the animation will be the lastFrameId plus one. Reference the animation by name.

```
void UChromaSDKPluginBPLibrary::TrimEndFramesName(const FString& animationName,
    int32 lastFrameId);
```

---

## TrimFrame

Remove the frame from the animation. Reference animation by id.

```
void UChromaSDKPluginBPLibrary::TrimFrame(int32 animationId, int32 frameId);
```

---

## TrimFrameName

Remove the frame from the animation. Reference animation by name.

```
void UChromaSDKPluginBPLibrary::TrimFrameName(const FString& animationName,
int32 frameId);
```

---

## TrimStartFrames

Trim the start of the animation starting at frame 0 for the number of frames. Reference the animation by id.

```
void UChromaSDKPluginBPLibrary::TrimStartFrames(int32 animationId, int32
numberOfFrames);
```

---

## TrimStartFramesName

Trim the start of the animation starting at frame 0 for the number of frames. Reference the animation by name.

```
void UChromaSDKPluginBPLibrary::TrimStartFramesName(const FString& animationName,
int32 numberOfFrames);
```

---

## UnloadAnimation

Unloads **Chroma** effects to free up resources. Returns the animation id upon success. Returns negative one upon failure. Reference the animation by id.

```
void UChromaSDKPluginBPLibrary::UnloadAnimation(const int32 animationId);
```

---

## UnloadAnimationName

Unload the animation effects. Reference the animation by name.

```
void UChromaSDKPluginBPLibrary::UnloadAnimationName(const FString& animationName);
```

---

### UseForwardChromaEvents

On by default, `UseForwardChromaEvents` sends the animation name to `CoreSetEventName` automatically when `PlayAnimationName` is called.

```
void UChromaSDKPluginBPLibrary::UseForwardChromaEvents(bool toggle);
```

---

### UseIdleAnimation

When the idle animation flag is true, when no other animations are playing, the idle animation will be used. The idle animation will not be affected by the API calls to `PluginIsPlaying`, `PluginStopAnimationType`, `PluginGetPlayingAnimationId`, and `PluginGetPlayingAnimationCount`. Then the idle animation flag is false, the idle animation is disabled. `Device` uses `EChromaSDKDeviceEnum` enums.

```
void UChromaSDKPluginBPLibrary::UseIdleAnimation(EChromaSDKDeviceEnum::Type device, bool flag);
```

---

### UseIdleAnimations

Set idle animation flag for all devices.

```
void UChromaSDKPluginBPLibrary::UseIdleAnimations(bool flag);
```

---

### UsePreloading

Set preloading animation flag, which is set to true by default. Reference animation by id.

```
void UChromaSDKPluginBPLibrary::UsePreloading(int32 animationId, bool flag);
```

---

### UsePreloadingName

Set preloading animation flag, which is set to true by default. Reference animation by name.

```
void UChromaSDKPluginBPLibrary::UsePreloadingName(const FString& animationName,  
    bool flag);
```

---